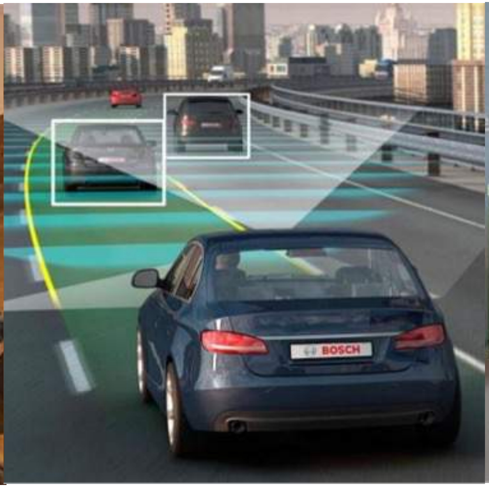
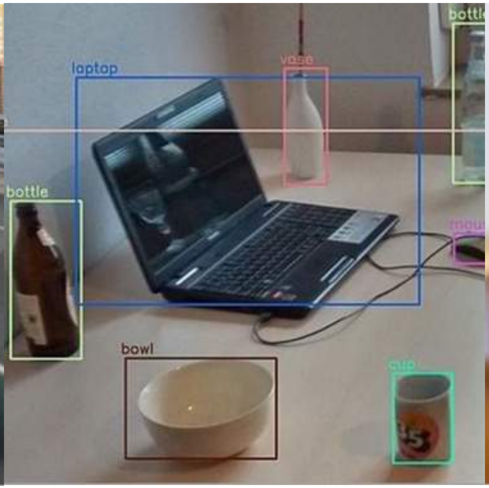
Introduction to Computer Vision





Lec.11: 3D Deep Learning

2023.12.11

Many slides adapted from “Tutoral on 3D Deep Learning ” of Hao Su (UCSD)

Today

Deep Learning for 3D reconstruction

1. Feature matching
2. Object Pose Estimation
3. Human Pose Estimation
4. Dense Reconstruction

Deep learning for 3D understanding

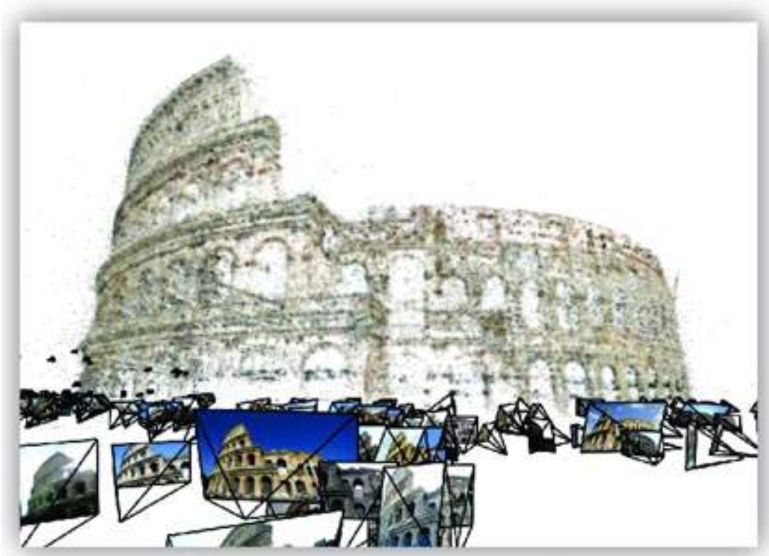
Today

Deep Learning for 3D reconstruction

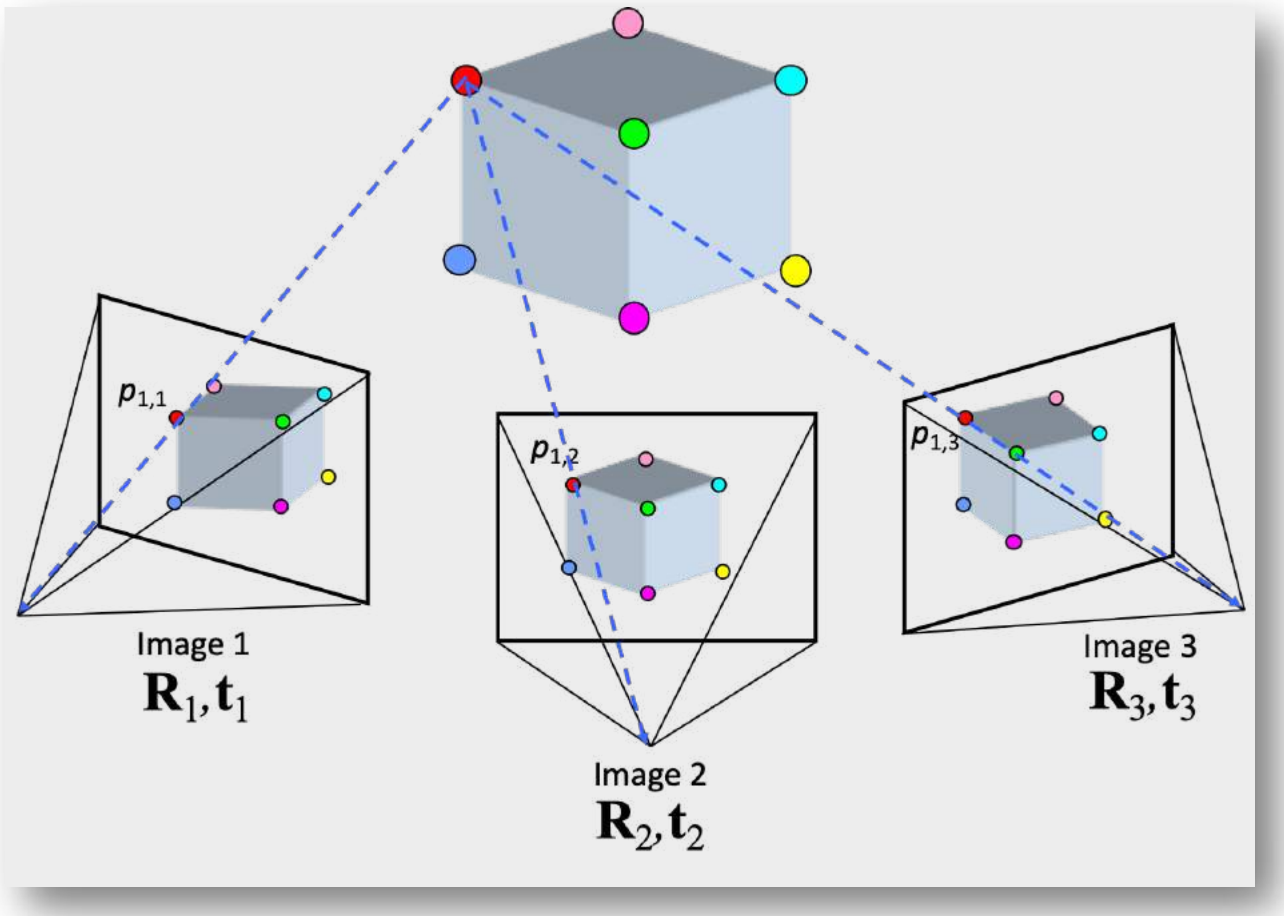
1. **Feature matching**
2. Object Pose Estimation
3. Human Pose Estimation
4. Dense Reconstruction

Deep learning for 3D understanding

Recap: SfM

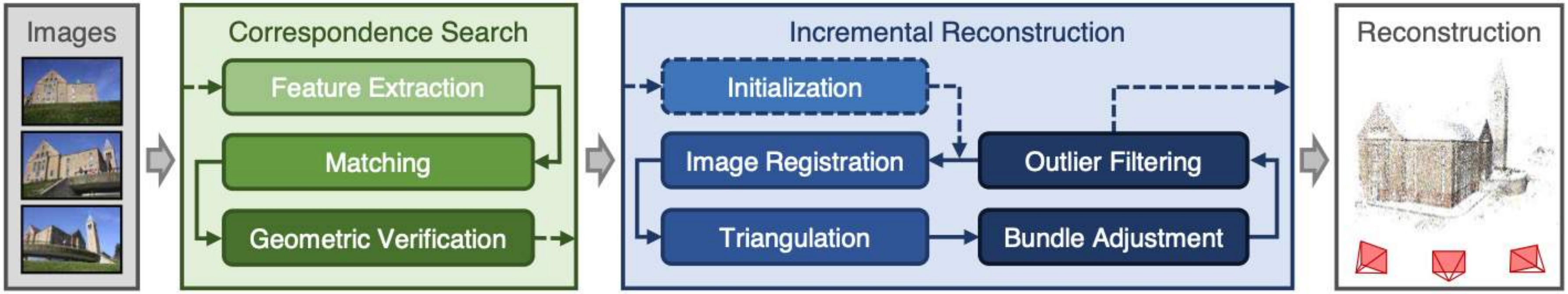


Recap: SfM



$p_{1,1}$
$p_{1,2}$
$p_{1,3}$
Image 1
$\mathbf{R}_1, \mathbf{t}_1$
Image 2
$\mathbf{R}_2, \mathbf{t}_2$
Image 3
$\mathbf{R}_3, \mathbf{t}_3$

Recap: Colmap

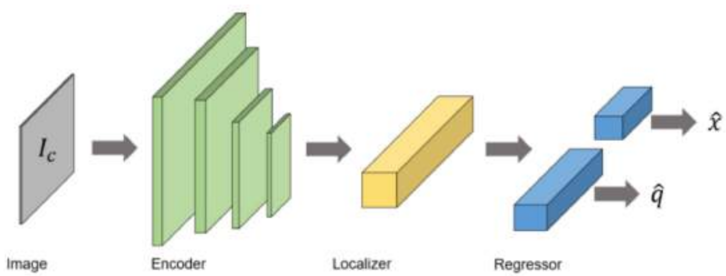


Learning to estimate pose

Using networks to directly predict pose from image?

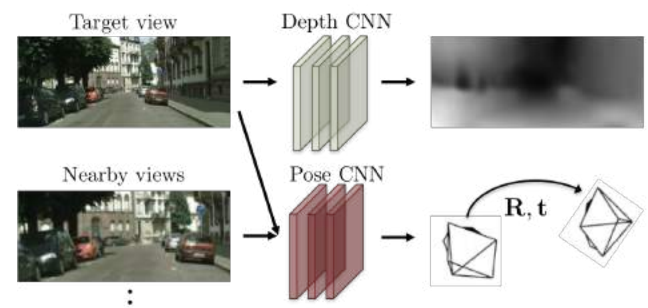
oseNet: A Convolutional Network for Real-Time 6-DOF Camera Relocalizatio

Alex Kendall Matthew Grimes Roberto Cipolla
University of Cambridge



Unsupervised Learning of Depth and Ego-Motion from Video

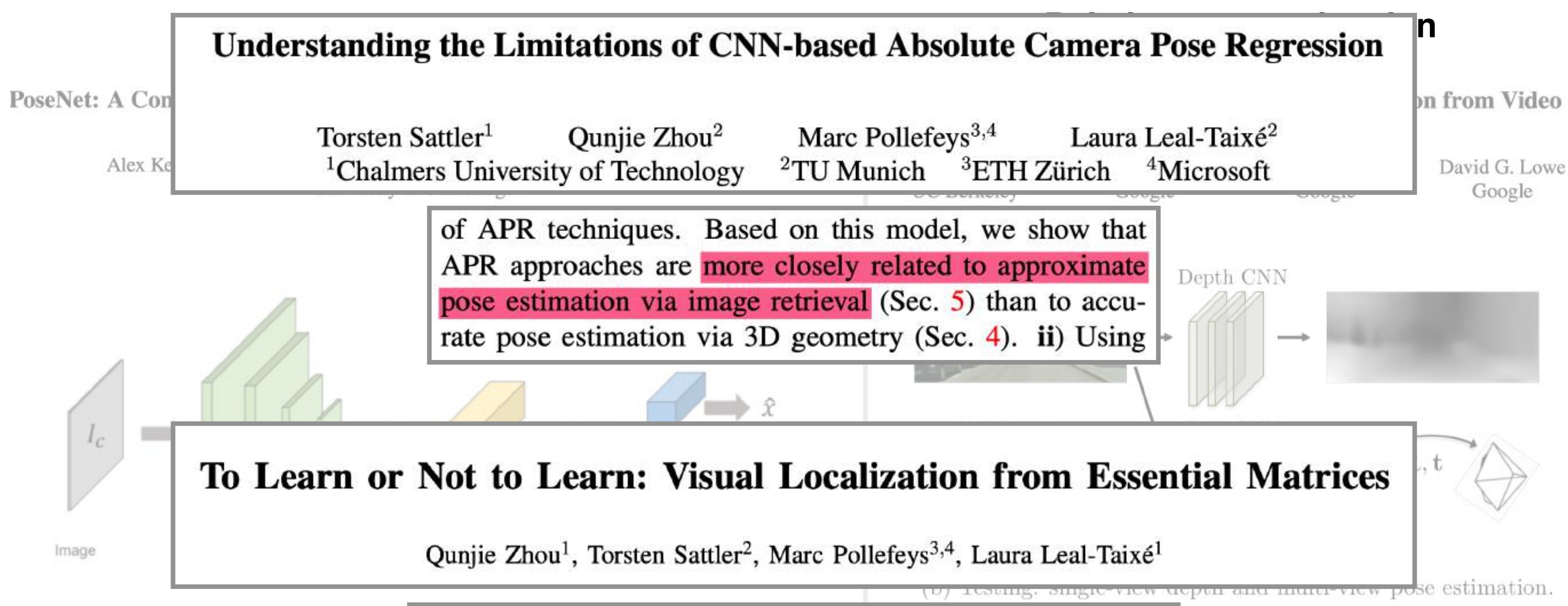
Tinghui Zhou* UC Berkeley
Matthew Brown Google
Noah Snavely Google
David G. Lowe Google



(b) Testing: single-view depth and multi-view pose estimation.

Learning to estimate pose

Using networks to directly predict pose is not good



Understanding the Limitations of CNN-based Absolute Camera Pose Regression

Torsten Sattler[1] Qunjie Zhou[2] Marc Pollefeys[3,4] Laura Leal-Taixé[2]
[1]Chalmers University of Technology [2]TU Munich [3]ETH Zürich [4]Microsoft

of APR techniques. Based on this model, we show that APR approaches are more closely related to approximate pose estimation via image retrieval (Sec. 5) than to accurate pose estimation via 3D geometry (Sec. 4). **ii**) Using

To Learn or Not to Learn: Visual Localization from Essential Matrices

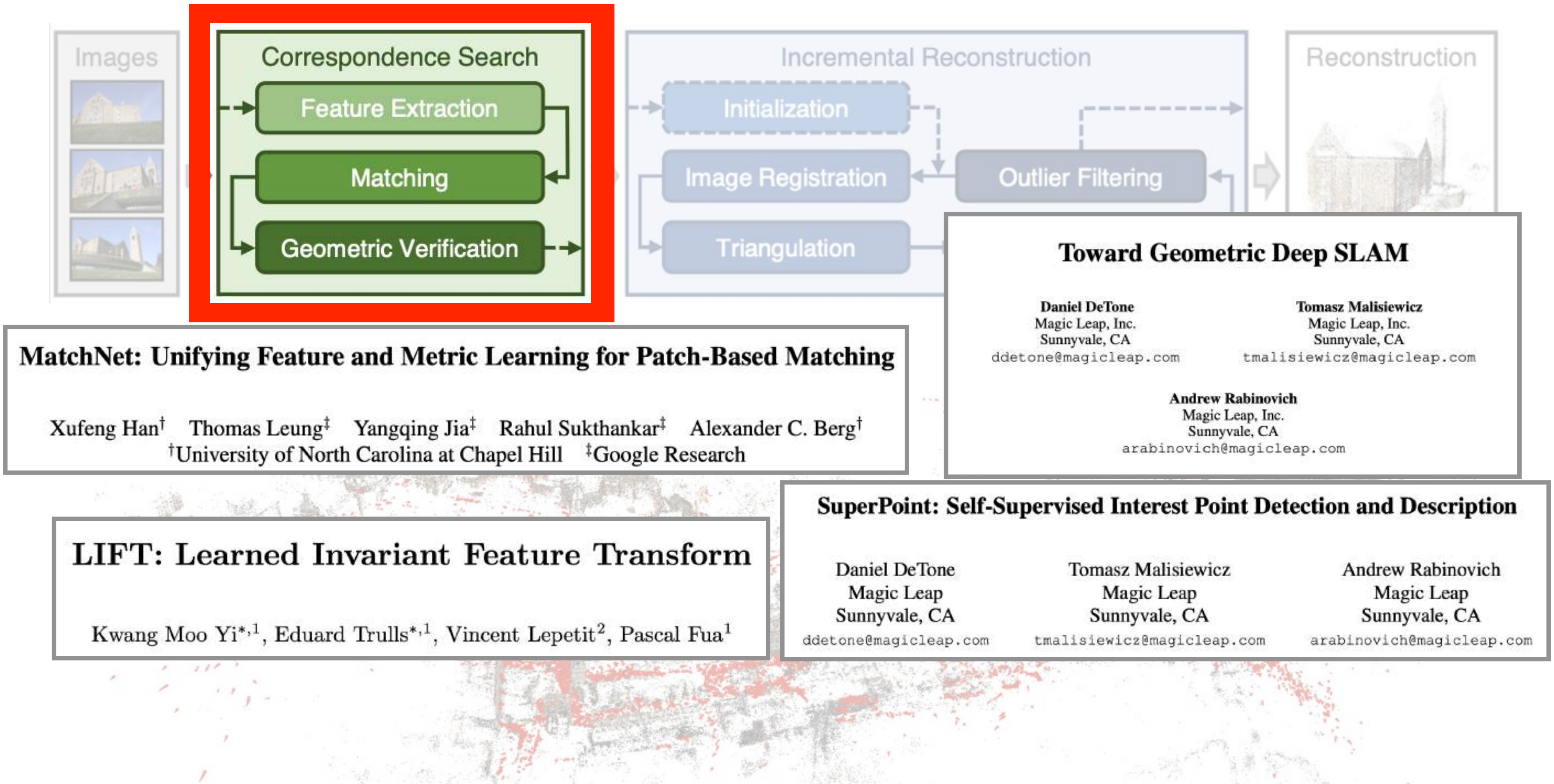
Qunjie Zhou[1], Torsten Sattler[2], Marc Pollefeys[3,4], Laura Leal-Taixé[1]

ing their results, we have shown that the purely data-driven approach does not generalize well and have identified the reason for this failure as the relative pose regression layers.

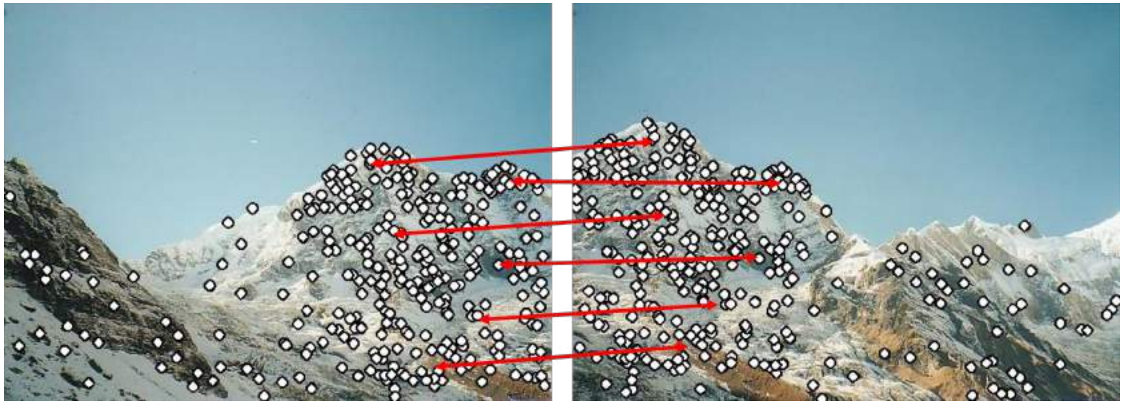
⋮

Learning to estimate pose

Use deep learning to improve feature matching



Recap: feature matching pipeline



Images | **Extract Features** | **Match Features**

1. Detection
2. Description

3. Matching

Why deep learning

Traditional feature detectors and descriptors are handcrafted (e.g. DoG, SIFT, ORB, …)

Limitations:

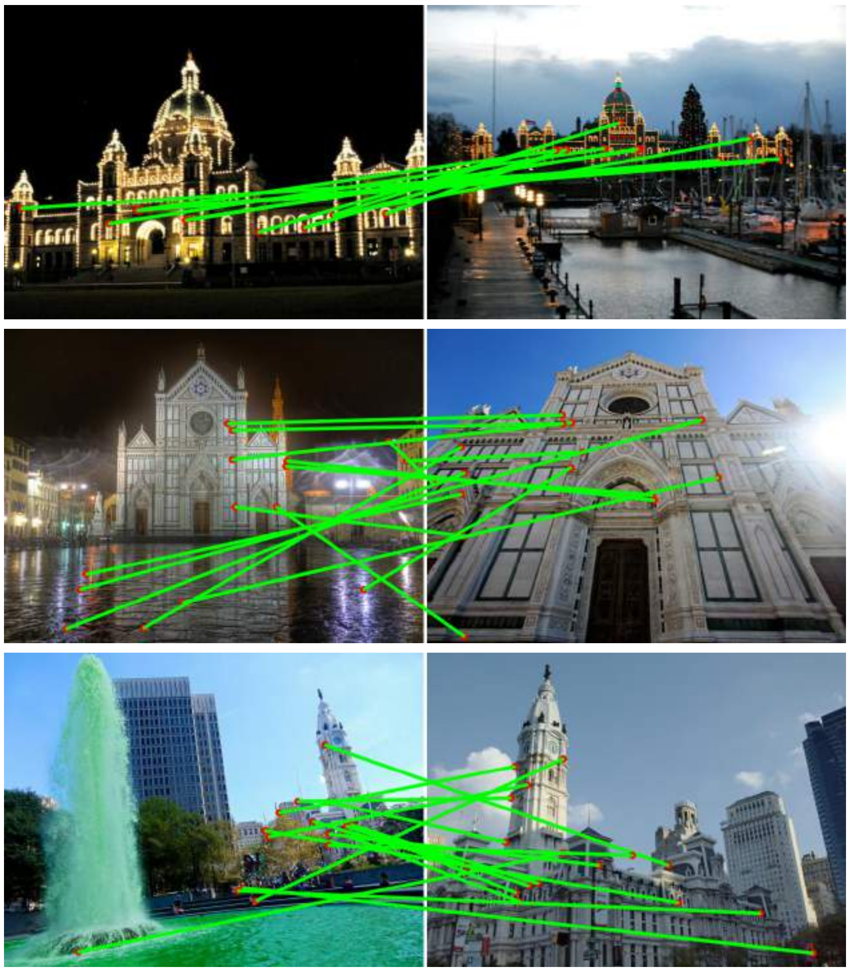
Geometry only, no semantics
Cannot handle poor texture
Not robust to

- viewpoint change
- illumination change
- motion blur

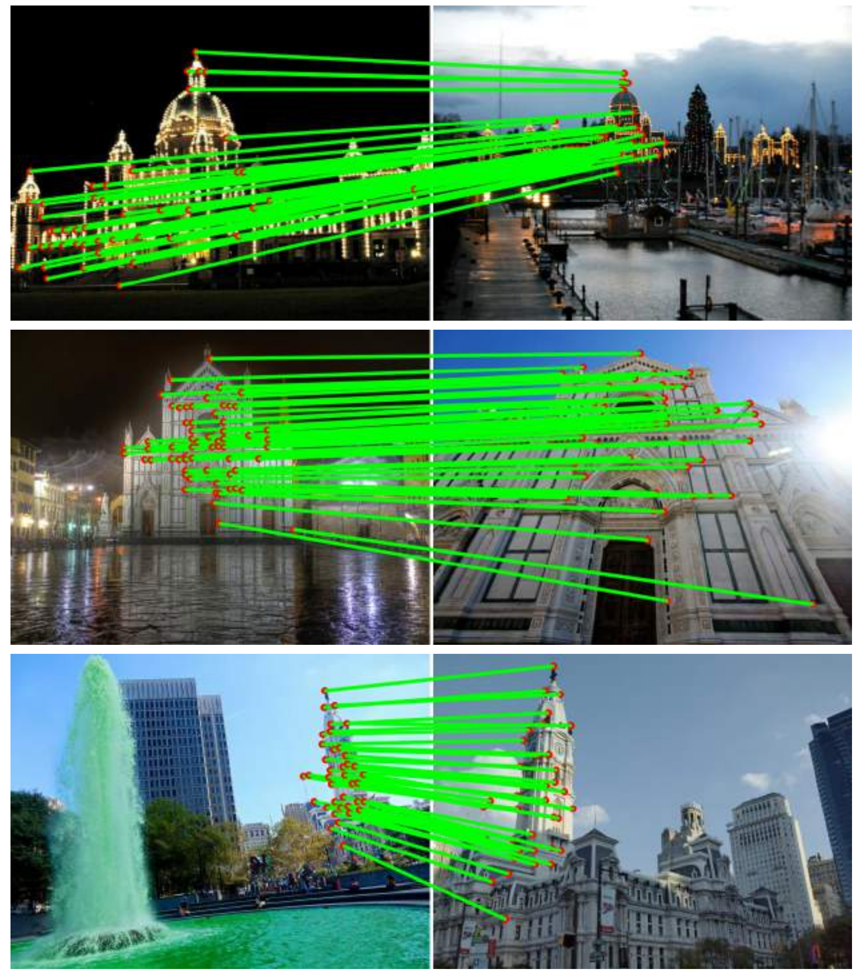
…

Deep learning for feature matching

SIFT

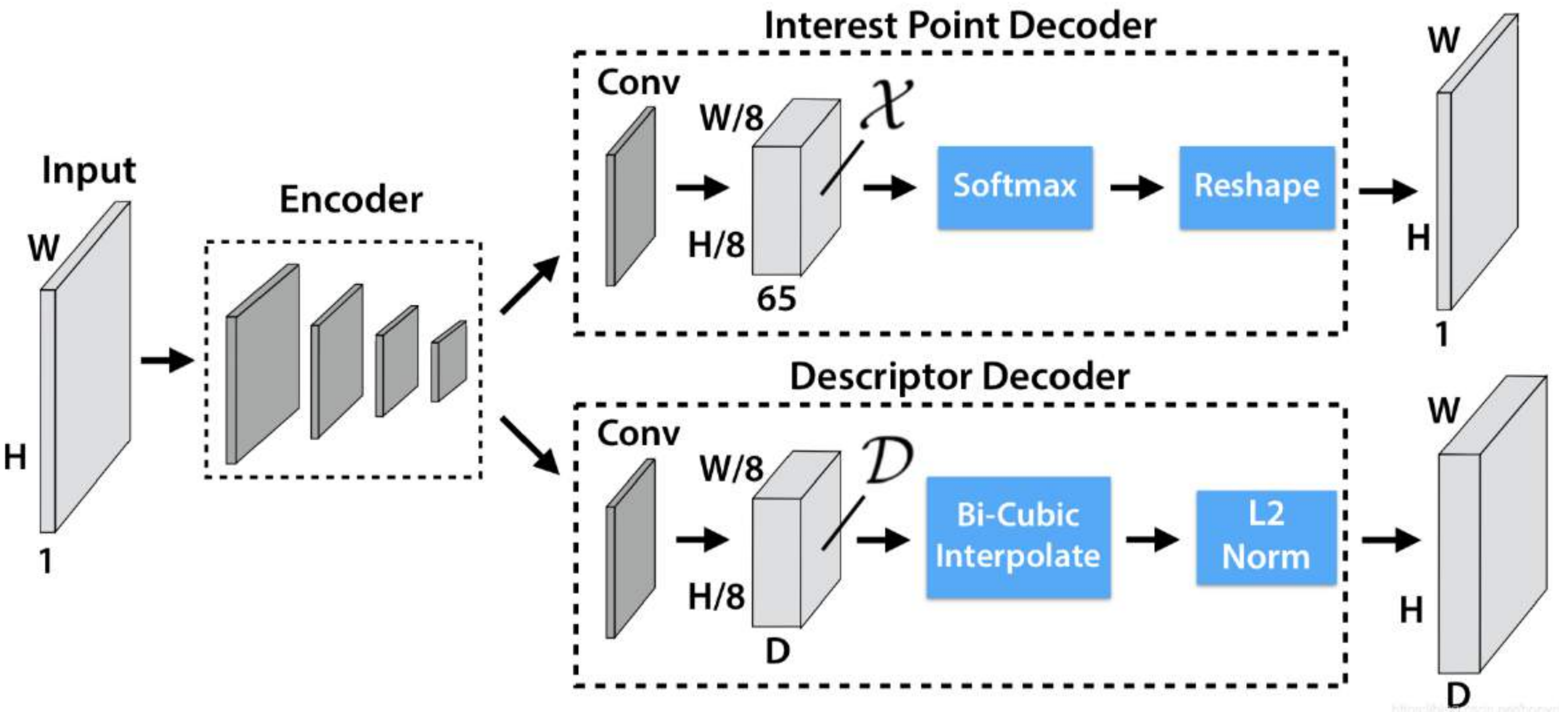


Learned



Deep learning for feature matching

Example: SuperPoint



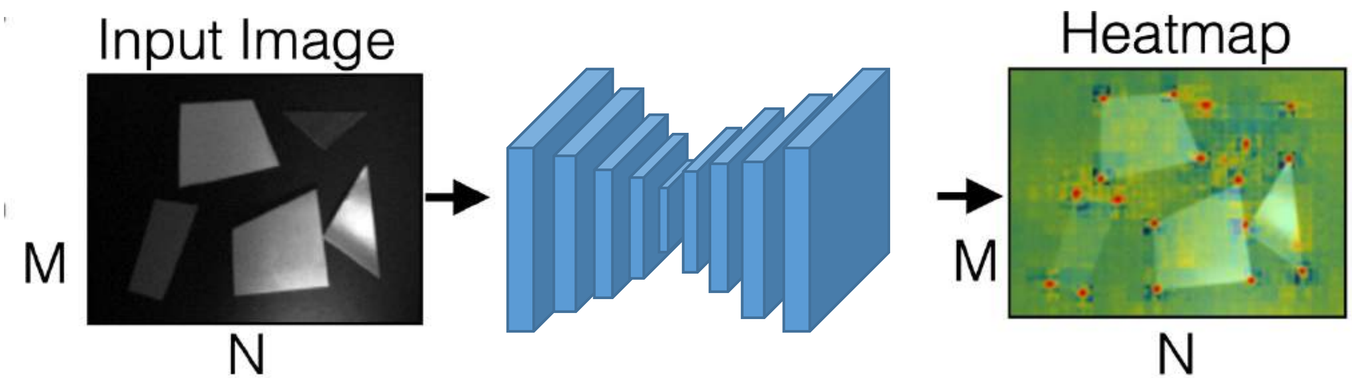
Interest points

Descriptors

DeTone, Daniel, Tomasz Malisiewicz, and Andrew Rabinovich. "Superpoint: Self-supervised interest point detection and description." *CVPR workshops*. 2018.

CNN-based detectors

Representing feature point locations by heatmaps

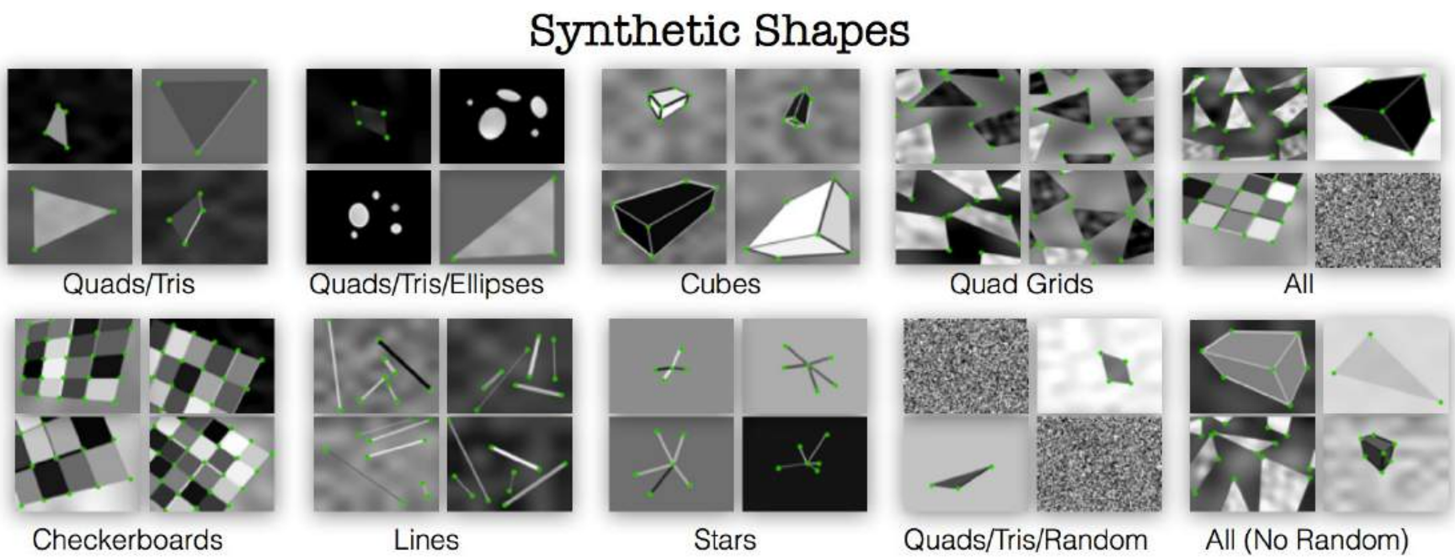


Fully ConvNet

NMS to get local maxima

Training detectors

Training CNNs to detect corners



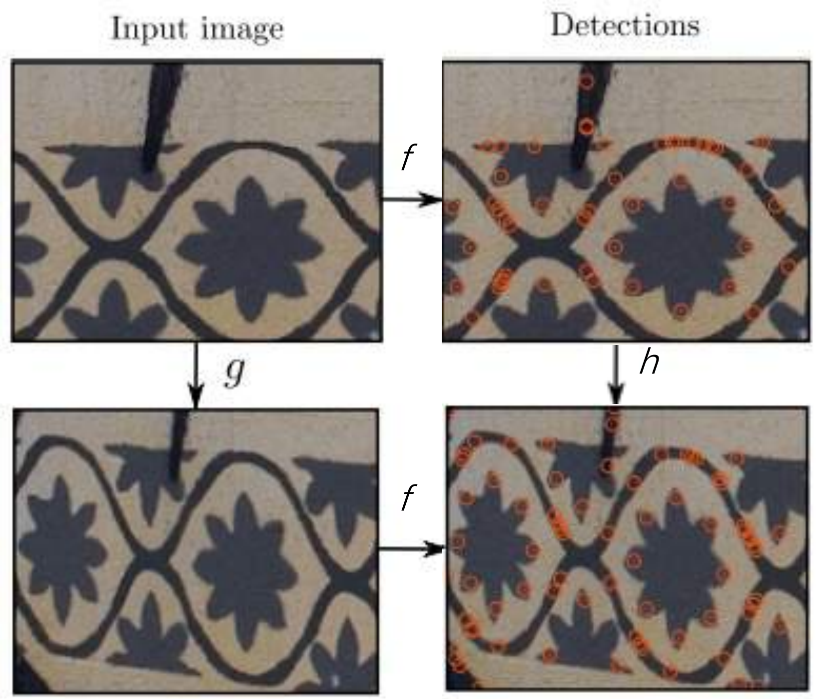
$$L_{loc} = \sum_i ||\mathbf{p}_t^* - \hat{\mathbf{p}}_t||_2 \ .$$

Training detectors

Training CNNs to enforce repeatability

1) Warp image
2) Enforce equivariance

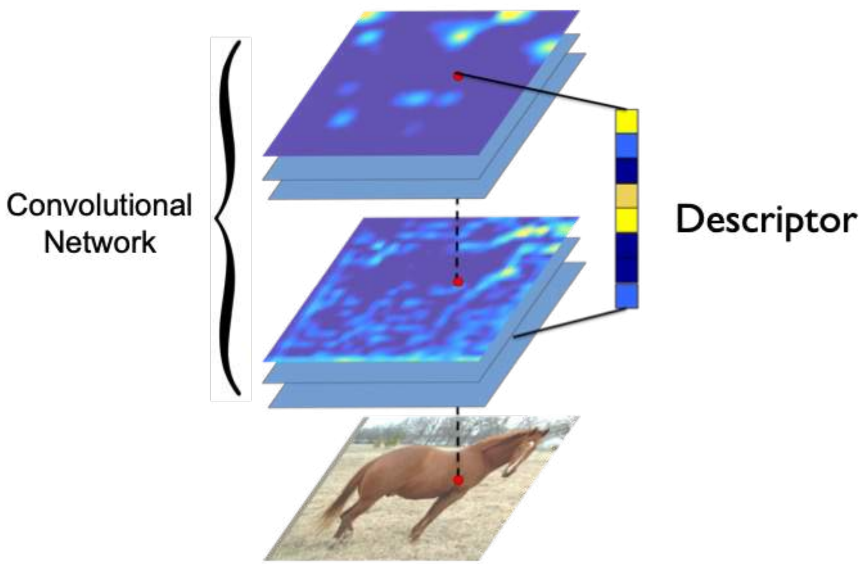
$$\min_f \frac{1}{n}\sum_{i=1}^{n}\left\|f\big(g(I)\big)-g\big(f(I)\big)\right\|^2$$



Lenc K, Vedaldi A. Learning covariant feature detectors. ECCV16

CNN-based descriptors

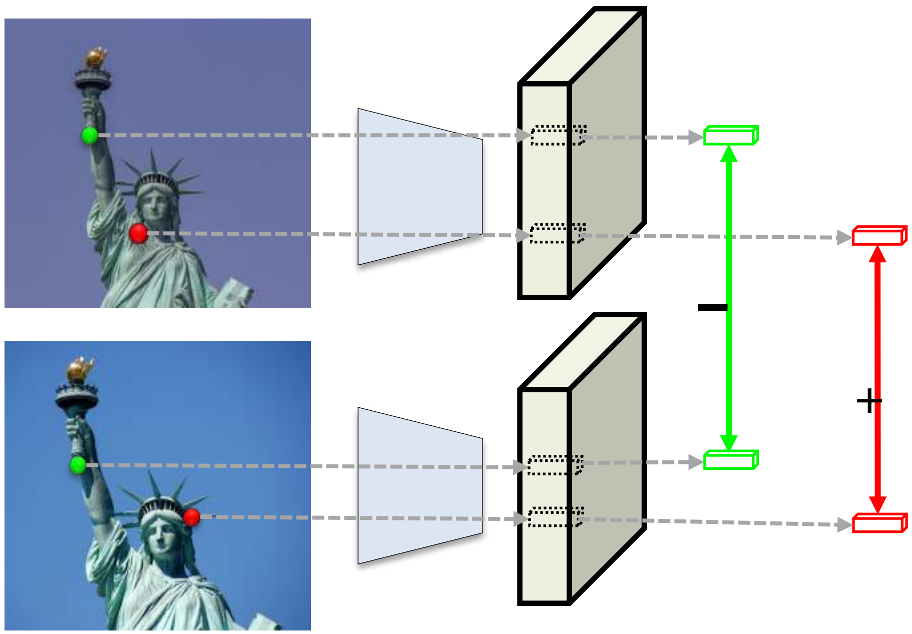
Extract descriptors from CNN feature maps



Hariharan, Bharath, et al. "Hypercolumns for object segmentation and fine-grained localization." *CVPR*. 2015.

Training descriptors

Training descriptors by metric learning

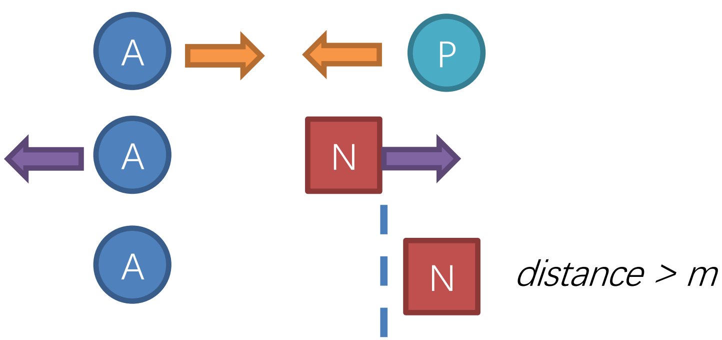


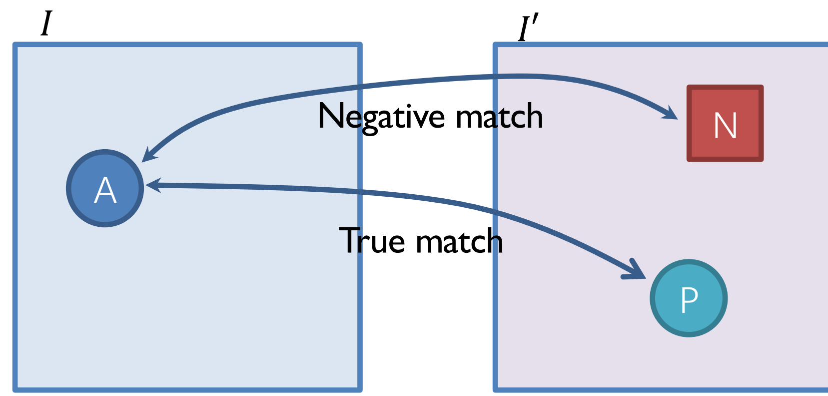
Training descriptors

Contrastive loss

$$L_{pos} = \frac{1}{N}\sum_{i=1}^{N} \|F_I(A) - F_{I'}(P)\|^2$$

$$L_{neg} = \frac{1}{N}\sum_{i=1}^{N} \max(0, m - \|F_I(A) - F_{I'}(N)\|)^2$$

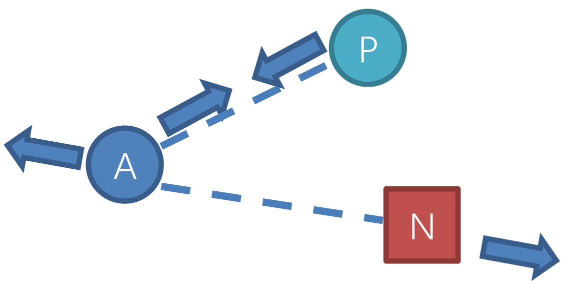




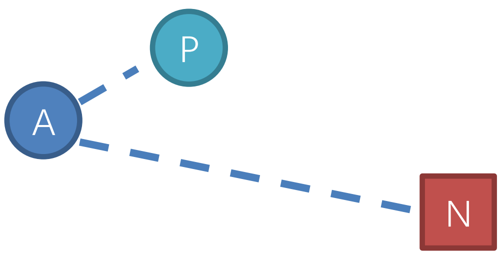
Training descriptors

Triplet loss

$$L_{tri} = \frac{1}{N}\sum_{i=1}^{N} \max(0, m + \|F_I(A) - F_{I'}(P)\| - \|F_I(A) - F_{I'}(N)\|)^2$$



metric distance < m

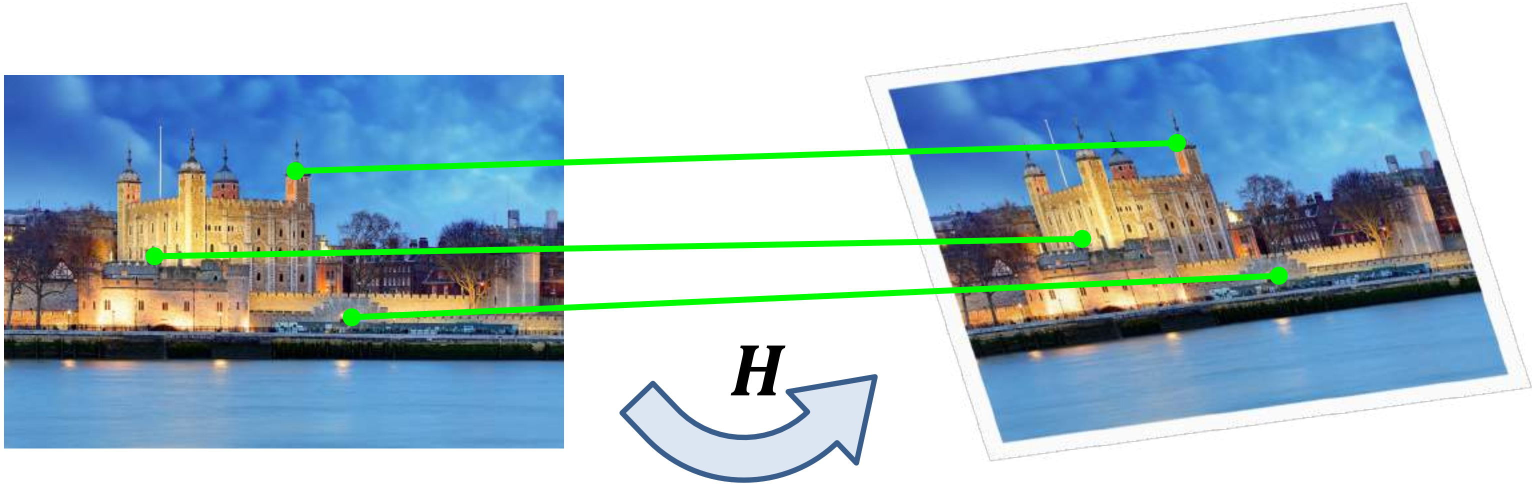


metric distance > m

Training descriptors

Where is training data from?

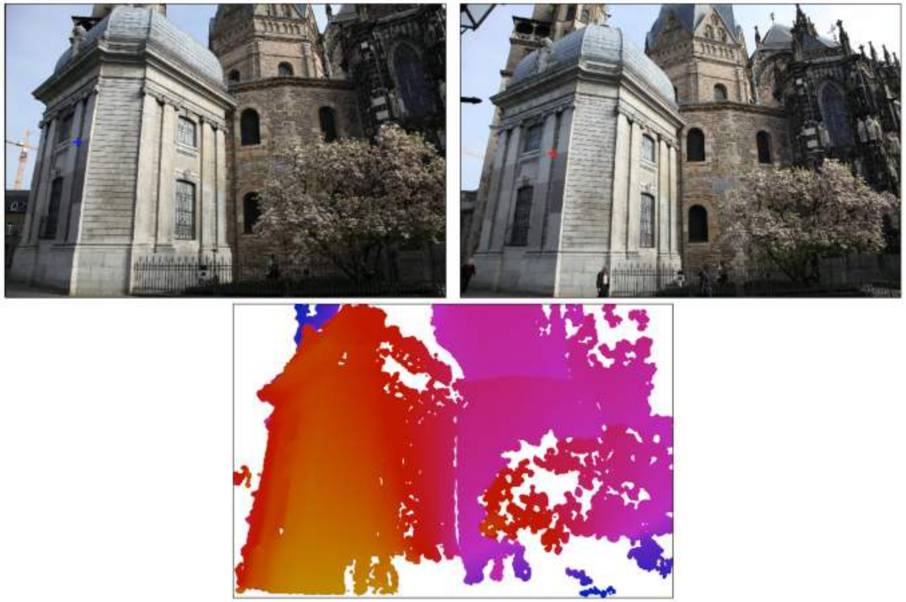
- An option is synthetic data



Training descriptors

Where is training data from?

- Another option is to use MVS



Questions?

Today

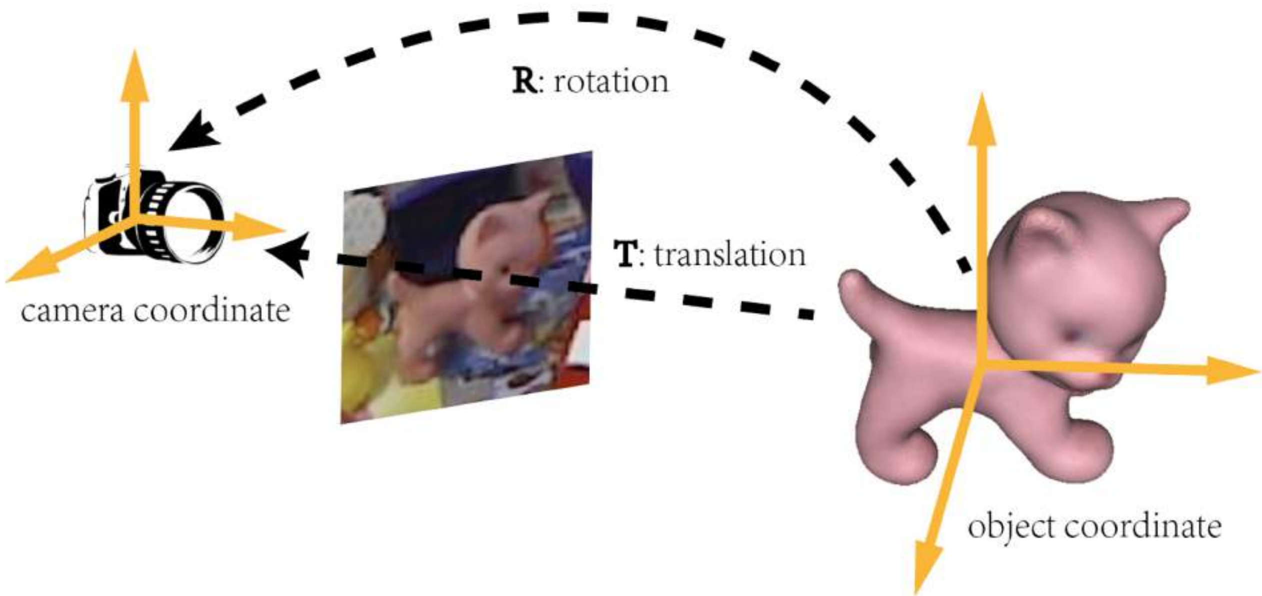
Deep Learning for 3D reconstruction

1. Feature matching
2. **Object Pose Estimation**
3. Human Pose Estimation
4. Dense Reconstruction

Deep learning for 3D understanding

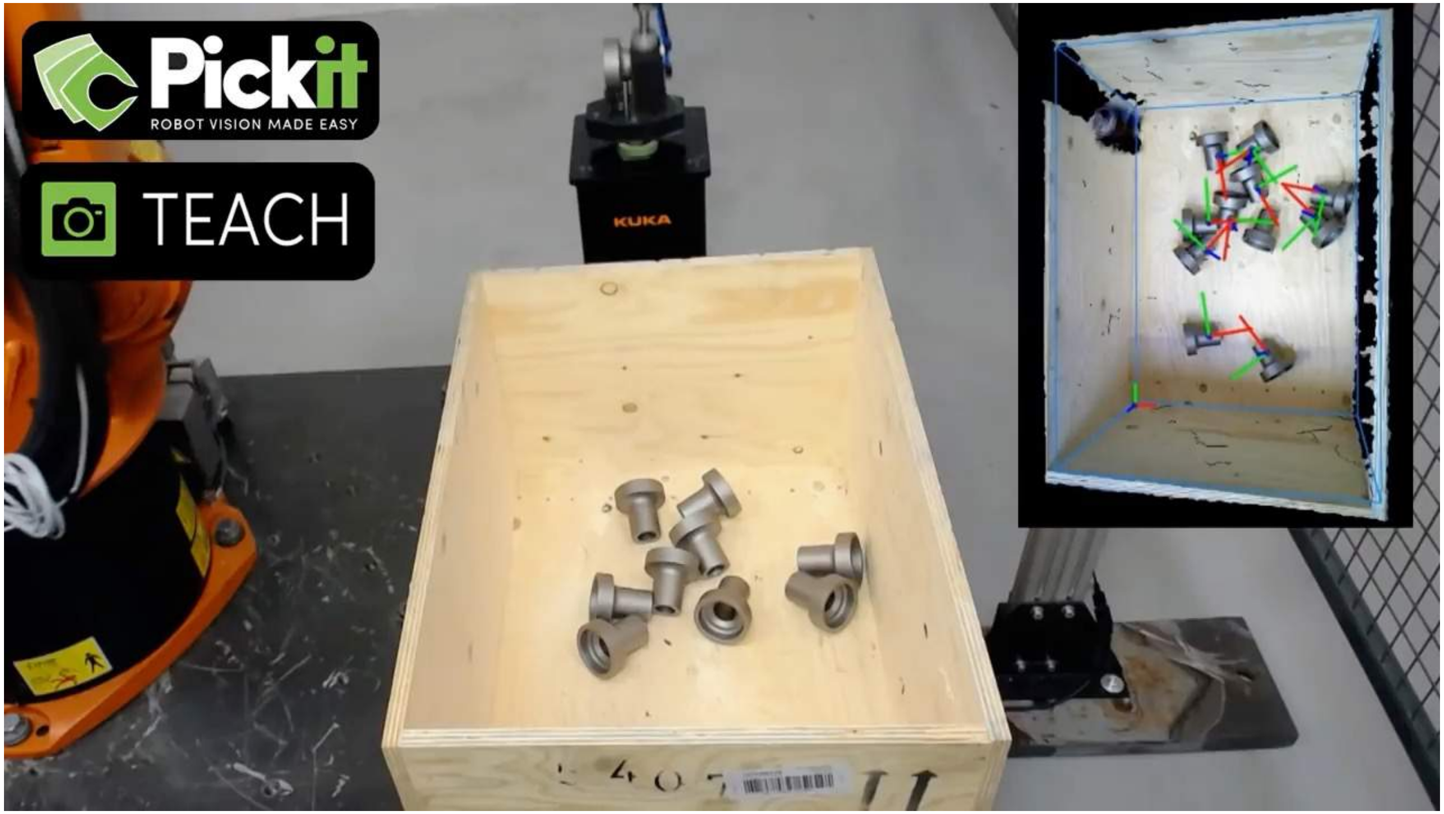
Object Pose Estimation

Estimate the **3D location and orientation** of an object realtive to the camera frame



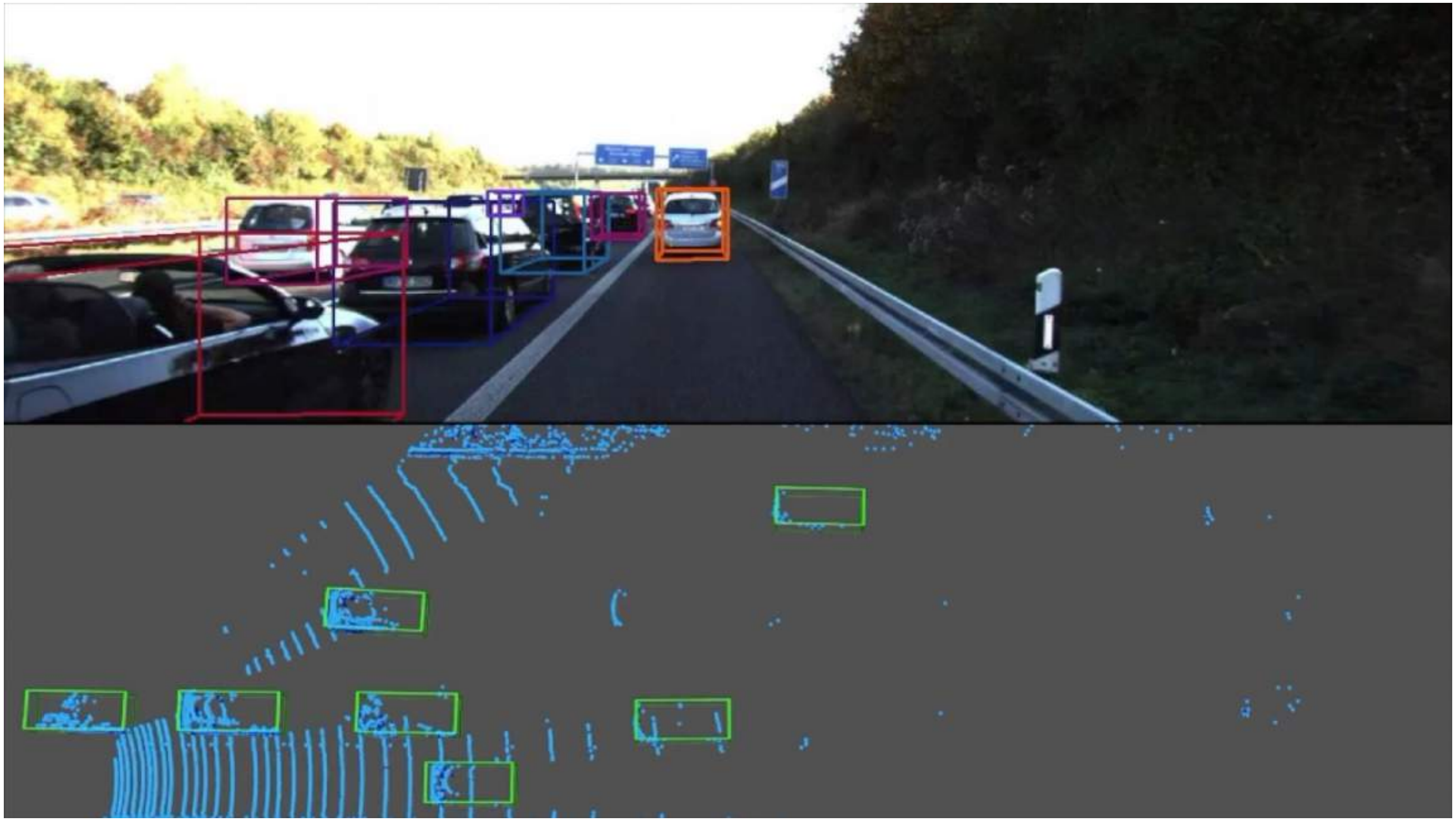
Object Pose Estimation

Applications: robot grasping



Object Pose Estimation

Applications: autonomous driving



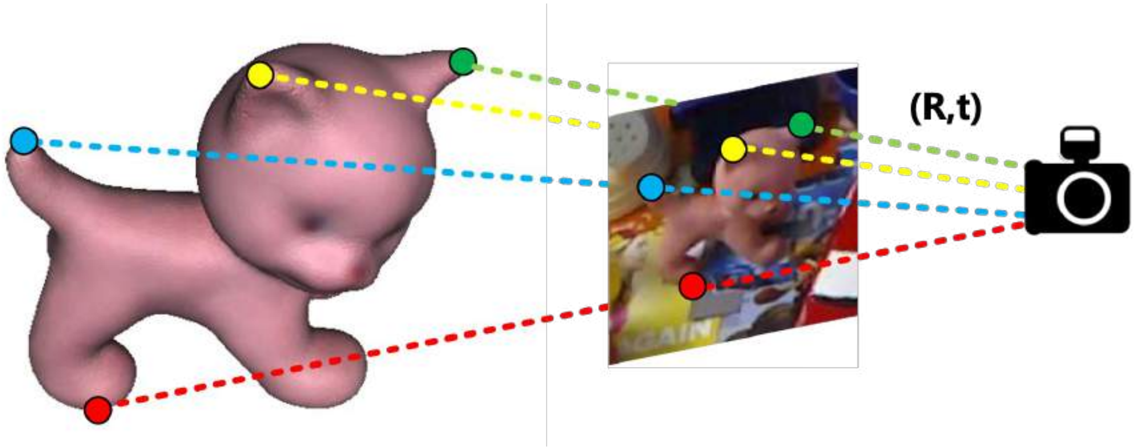
Object Pose Estimation

Applications: augmented reality

Object Pose Estimation

Similar to visual localization:

1. Find 3D-2D correspondences
2. Solve ***R*** and ***t*** by perspective-n-point (PnP) algorithm



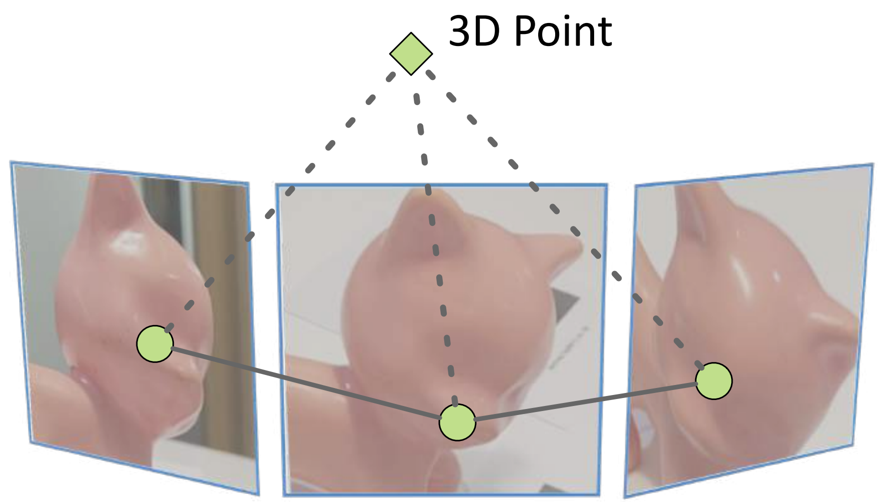
How to find 3D-2D correspondences?

Object Pose Estimation

Feature-matching-based methods

- First, reconstruct object SfM model by input multi-view images

Input Multi-View Reference Images

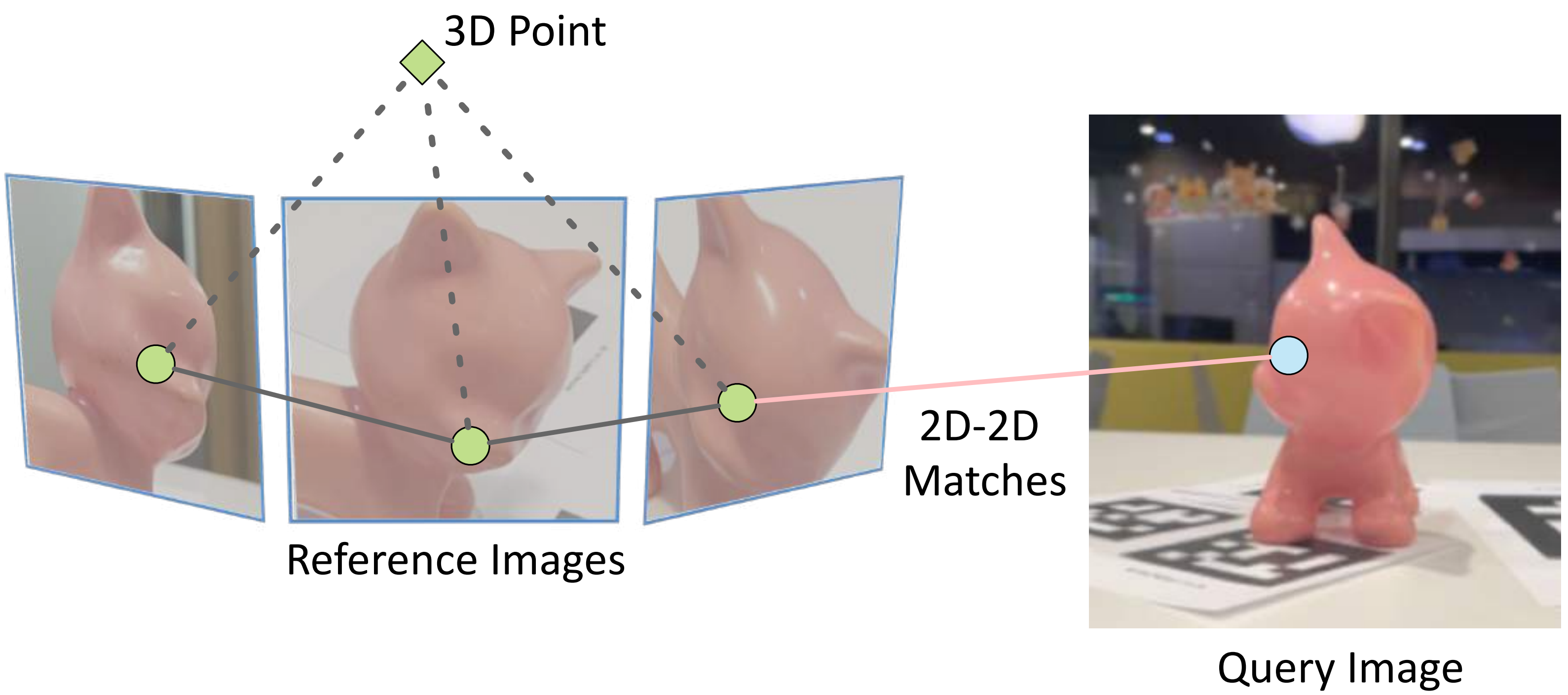


Structure from Motion

Object Pose Estimation

Feature-matching-based methods

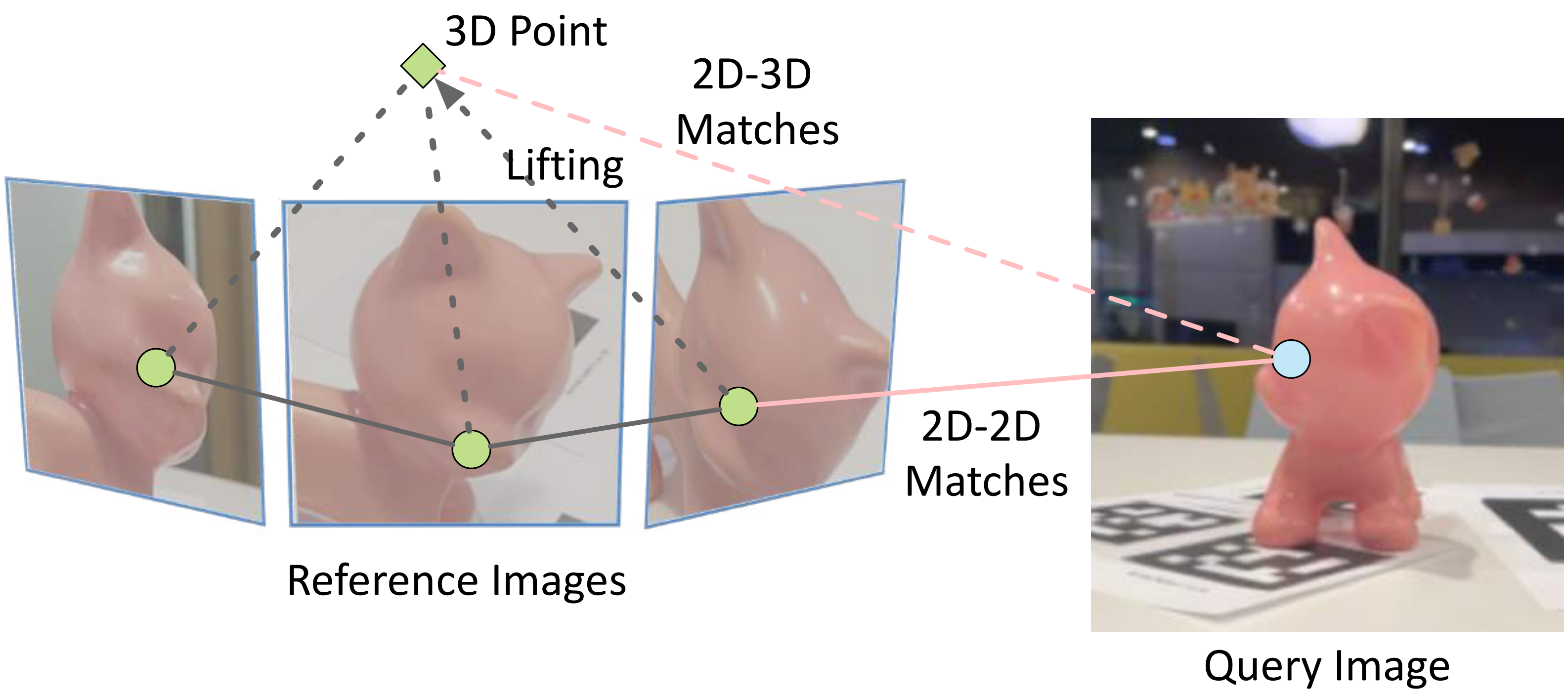
- Then obtain 2D-3D correspondeces by lifting 2D-2D matches to 3D



Object Pose Estimation

Feature-matching-based methods

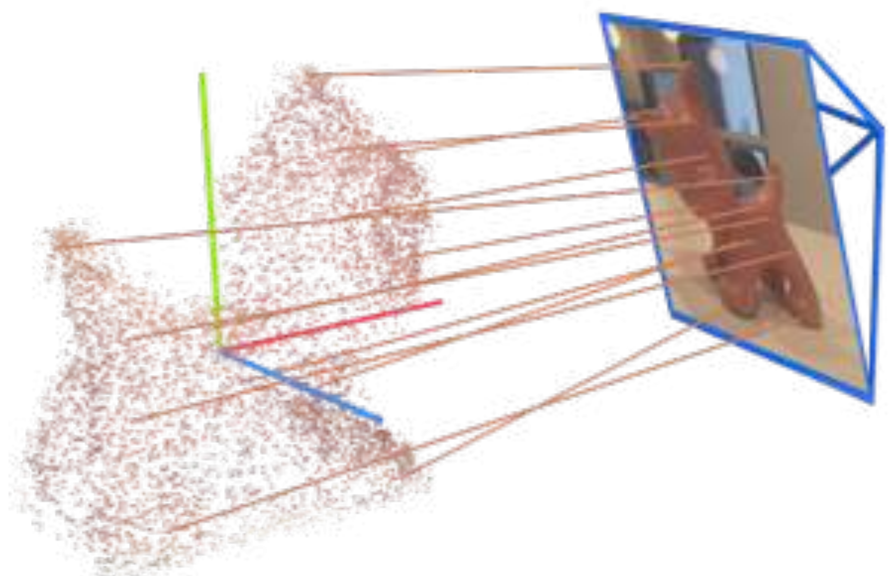
- Then obtain 2D-3D correspondeces by lifting 2D-2D matches to 3D



Object Pose Estimation

Feature-matching-based methods

- Finally, object pose of query image can be solved by PnP

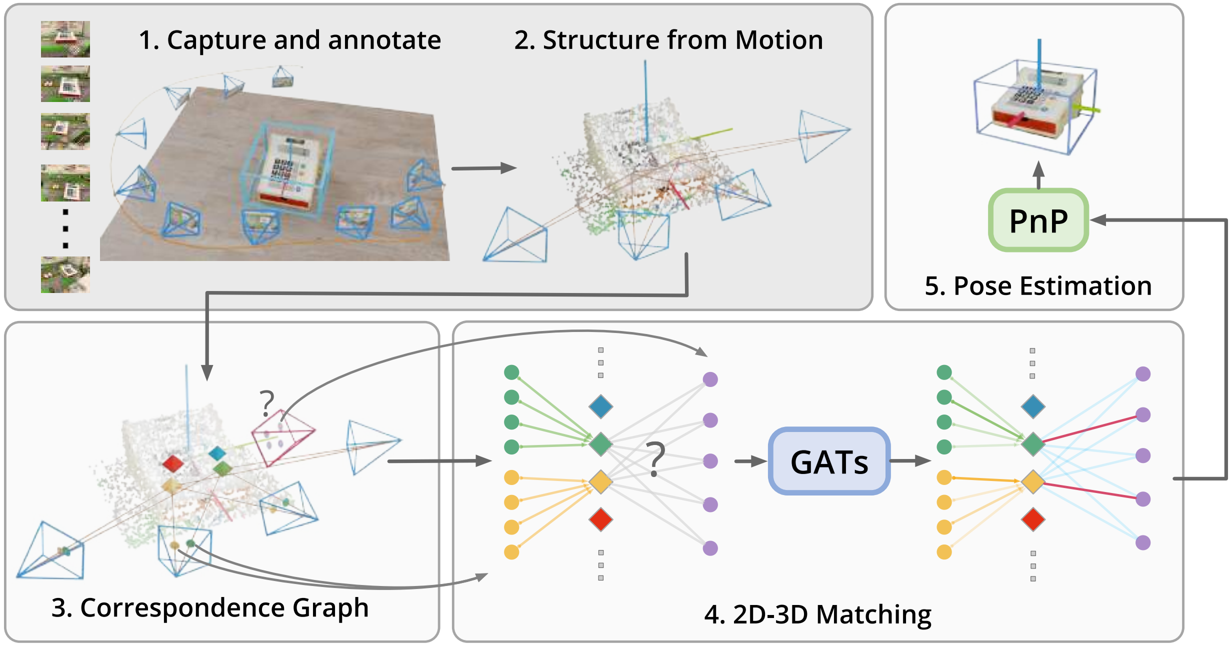


2D-3D correspondeces between query image and SfM model

Object Pose Estimation

Feature-matching-based methods

- Futher reading list

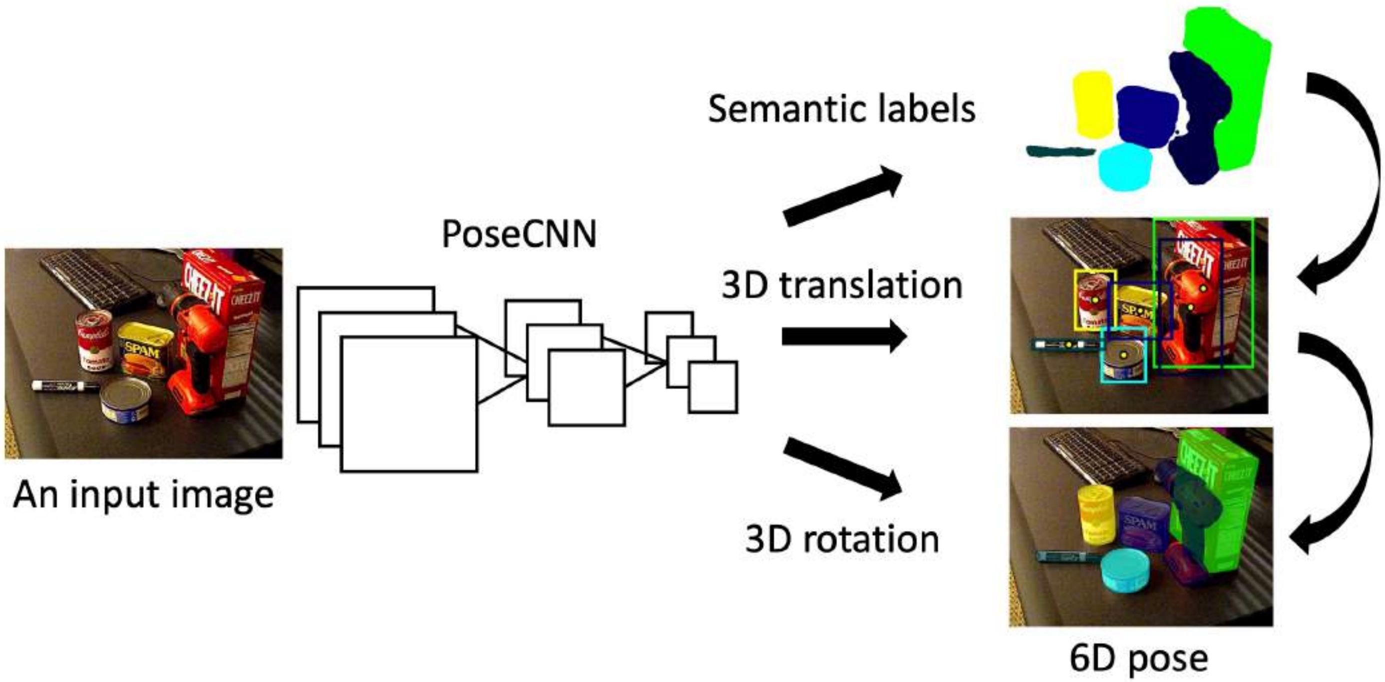


Skrypnyk, Iryna and David G. Lowe. “Scene modelling, recognition and tracking with invariant image features.” *ISMAR* (2004)
Sun, Jiaming et al. “OnePose: One-Shot Object Pose Estimation without CAD Models.” *CVPR* (2022)
Liu, Yuan et al. “Gen6D: Generalizable Model-Free 6-DoF Object Pose Estimation from RGB Images.” *ECCV* (2022).

Object Pose Estimation

Direct Pose Regression Methods

- Directly regressing object pose of query image using a neural network
- Need to render a large amount of images for training

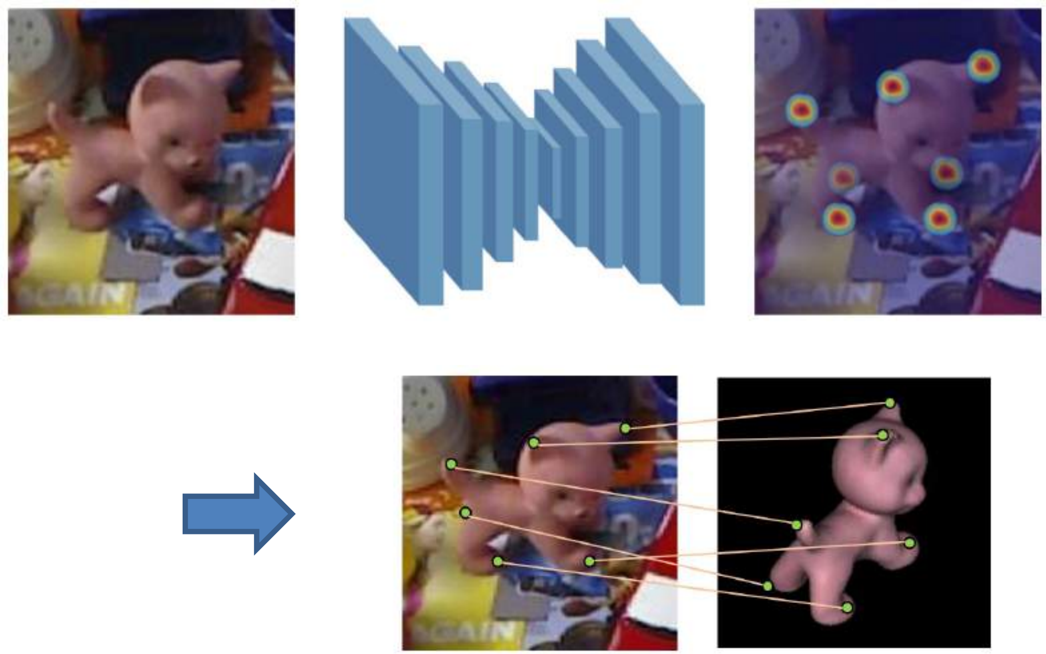


Xiang, Yu et al. "PoseCNN: A Convolutional Neural Network for 6D Object Pose Estimation in Cluttered Scenes."

Object Pose Estimation

Keypoint detection methods

- Using a CNN to detect pre-defined keypoints
- Need to render a large amount of images for training



Pavlakos et al. “6DoF Object Pose from Semantic Keypoints”, ICRA 2017.
Peng, Sida et al. “PVNet: Pixel-Wise Voting Network for 6DoF Pose Estimation.” *(CVPR)* (2019)

Today

Deep Learning for 3D reconstruction

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3. **Human Pose Estimation**
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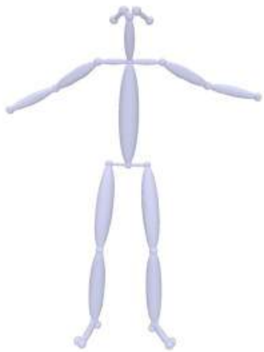
Deep learning for 3D understanding

3D Human Pose Estimation

2D Human Pose Estimation:
Localize human joints (keypoints – elbows, wrists, etc) in images.

3D Human Pose Estimation:
Estimate a 3D (x, y, z) coordinate for each joint from image

Input video



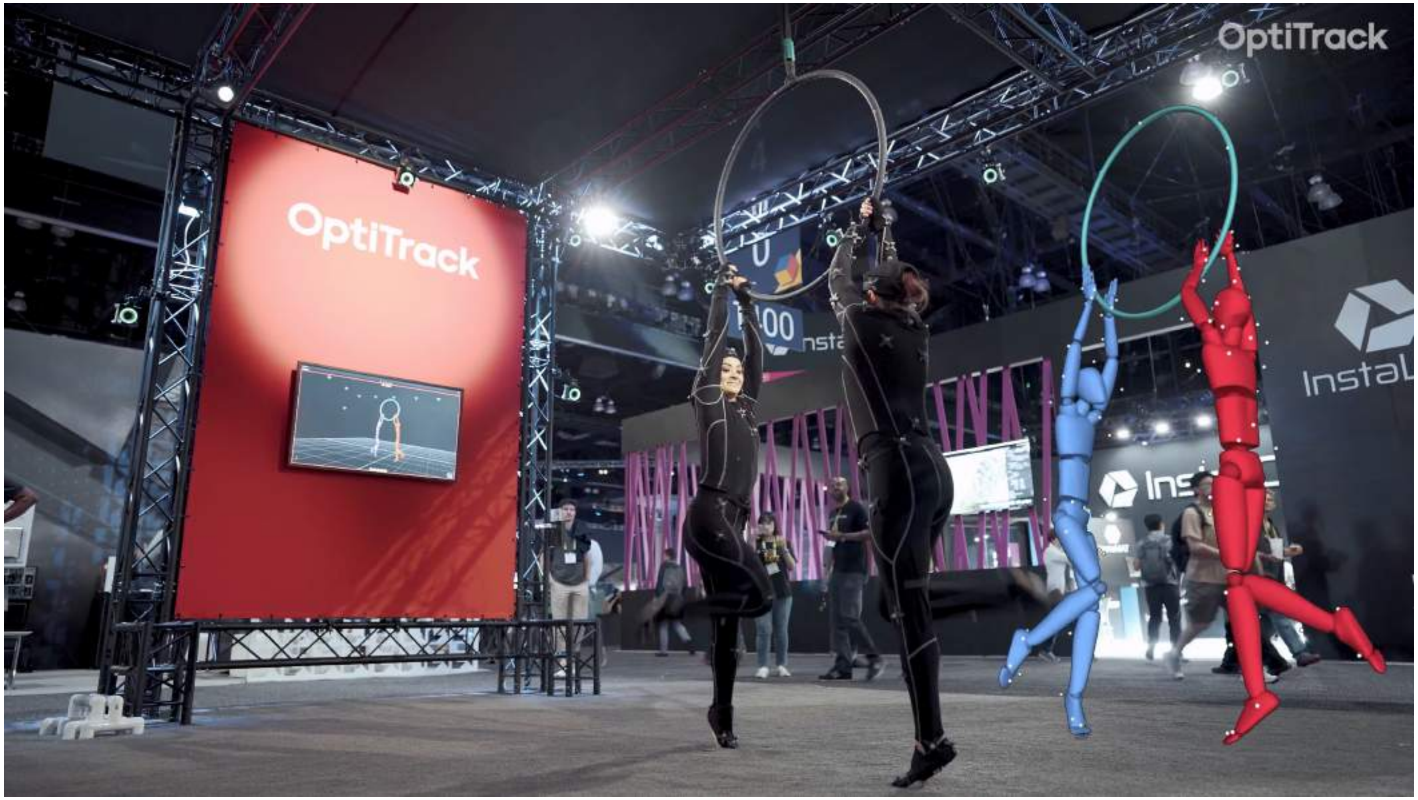
Skeleton

3D Human Pose Estimation

https://www.dailymail.co.uk/

Marker-based MoCap system

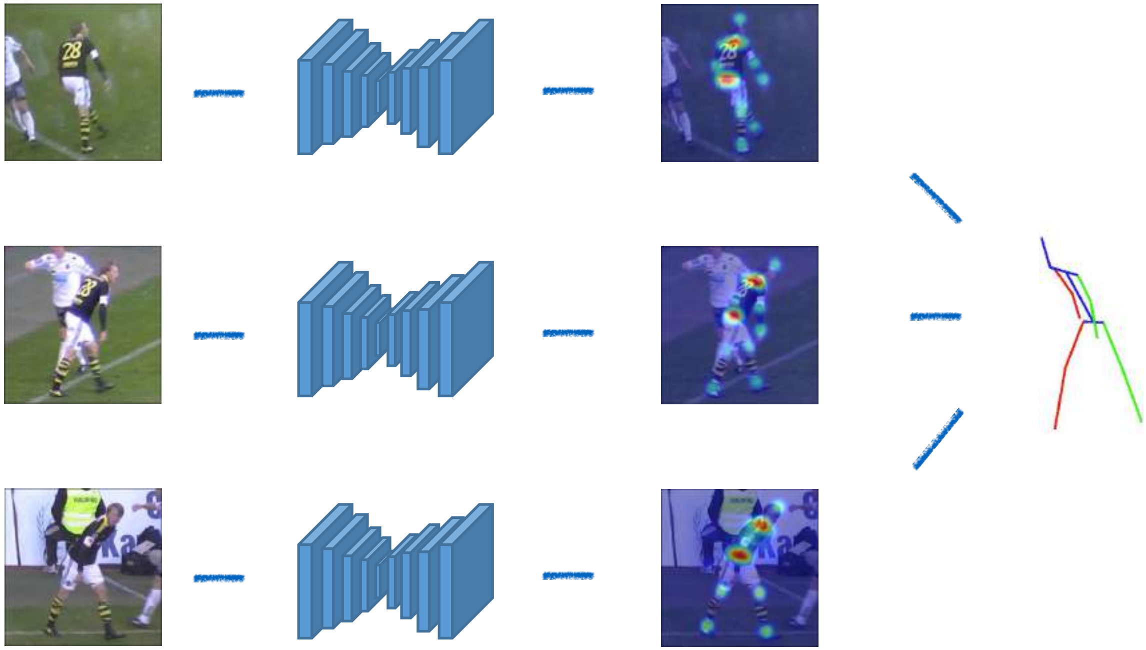
Optical motion capture (MoCap) system



https://www.youtube.com/watch?v=x-nIJ0mYghQ

Markerless MoCap

Multiview 3D Human Pose Estimation



Harvesting Multiple Views for Marker-less 3D Human Pose Annotations. CVPR 2017

Markerless MoCap

"Fast and robust multi-person 3d pose estimation from multiple views." *CVPR*. 2019.

Monocular 3D Human Pose Estimation

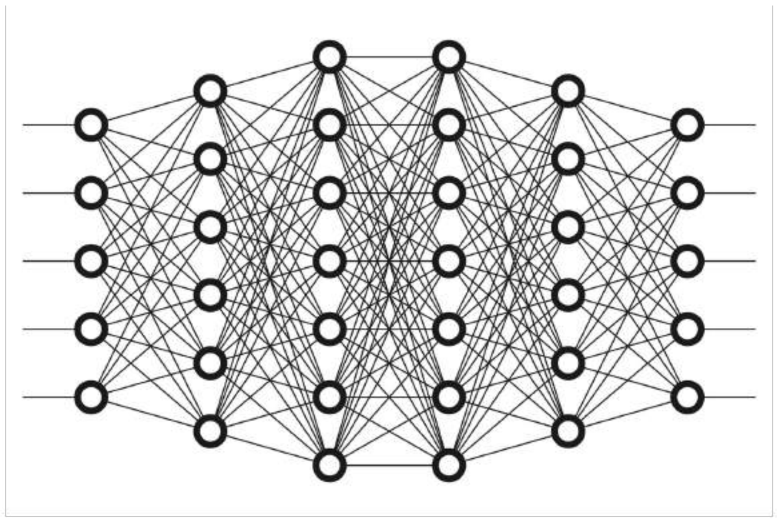
Estimating 3D human pose using a single camera

Mehta, Dushyant, et al. "Vnect: Real-time 3d human pose estimation with a single rgb camera." *ACM Transactions on Graphics (TOG)* 36.4 (2017): 1-14.

Monocular 3D Human Pose Estimation

Estimating 3D human pose from a single RGB image

Input image



$$\begin{matrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \\ \vdots & \vdots & \vdots \\ x_N & y_N & z_N \end{matrix}$$

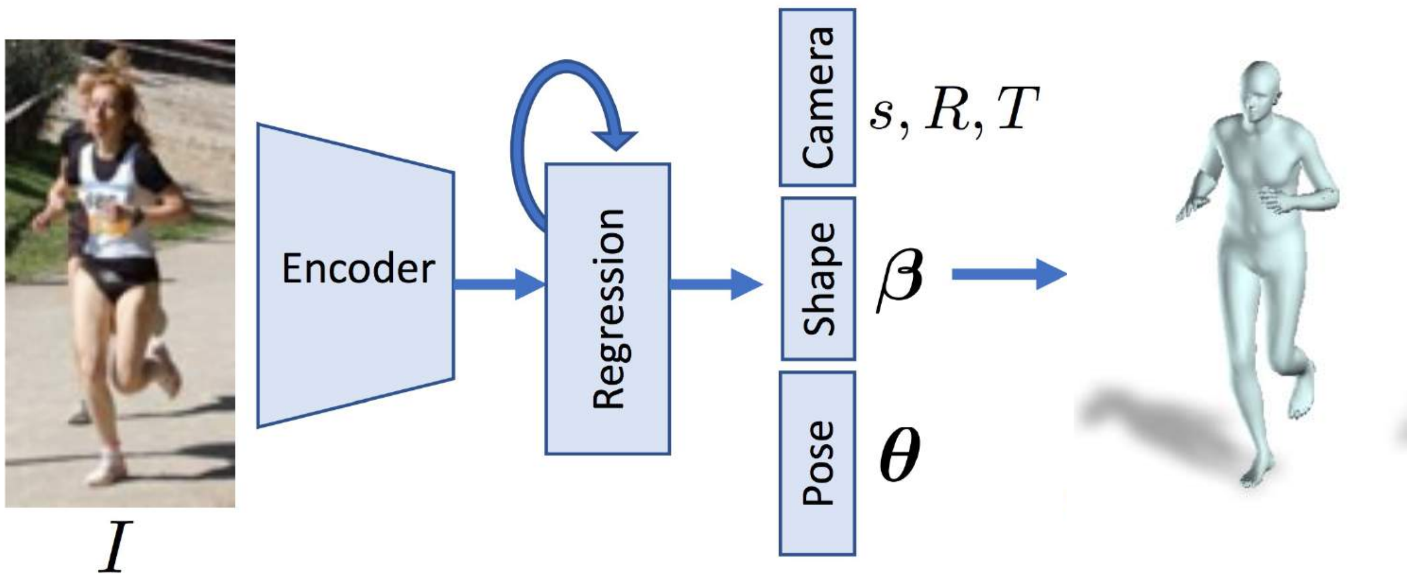
Joint locations

Monocular 3D Human Pose Estimation

Estimating 3D human pose from a single RGB image

Monocular 3D Human Pose Estimation

Estimating 3D human pose from a single RGB image



Loper et al. “SMPL: A Skinned Multi-Person Linear Model”. In SIGGRAPH Asia, 2016

Pavlakos et al. “Expressive Body Capture: Hands, Face, and Body from a Single Image”. CVPR, 2019.

Monocular 3D Human Pose Estimation

Estimating 3D human pose from a single RGB image

https://github.com/mkocabas/VIBE

Today

Deep Learning for 3D reconstruction

1. Feature matching
2. Object Pose Estimation
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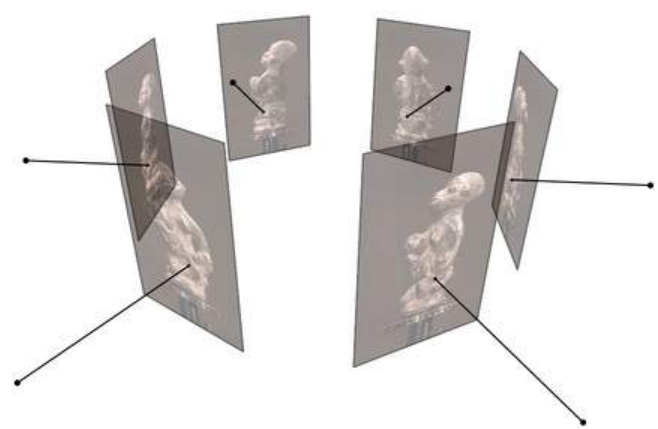
4. **Dense Reconstruction**

Deep learning for 3D understanding

Dense Reconstruction

Traditional pipeline:

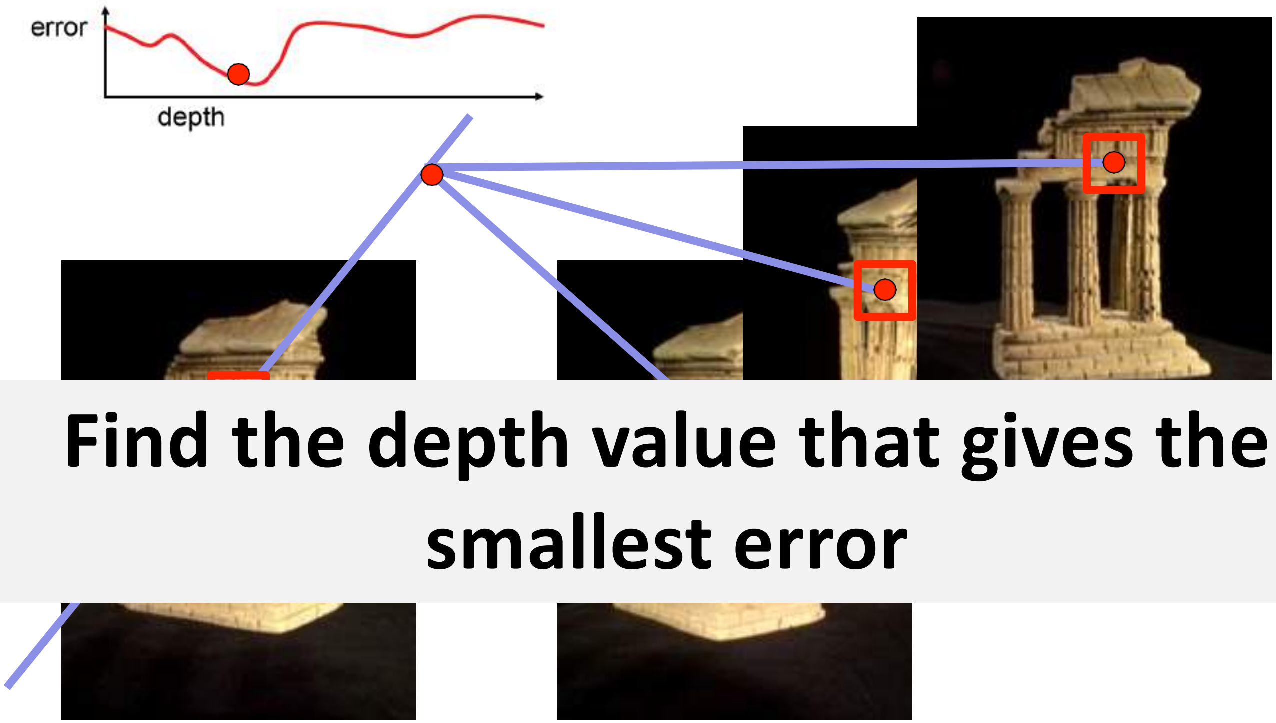
- Compute depth map per image (multi-view stereo)
- Fuse the depth maps into a 3D surface (Poisson reconstruction)
- Texture mapping



Figures by Carlos Hernandez

Recap: multi-view stereo

Compute the error for each depth value for each point in the reference image



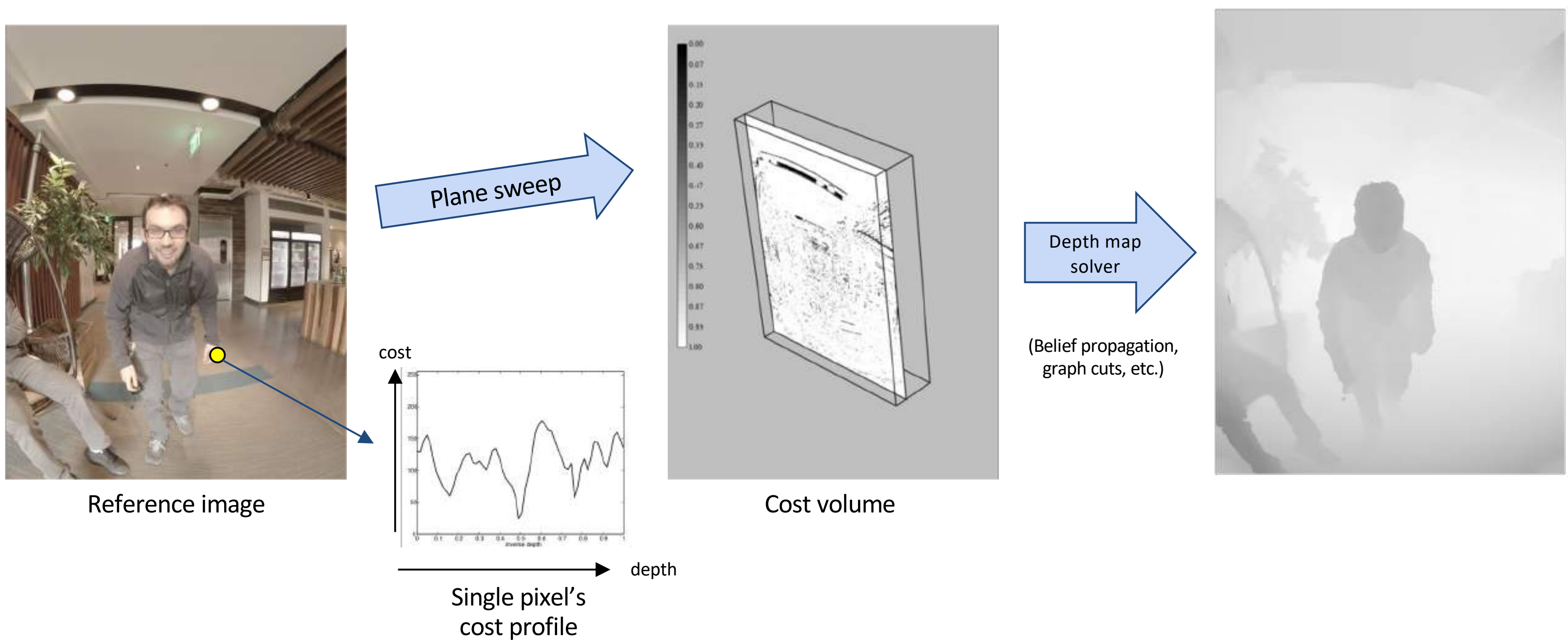
Find the depth value that gives the smallest error

reference view

neighbor views

Source: Y. Furukawa

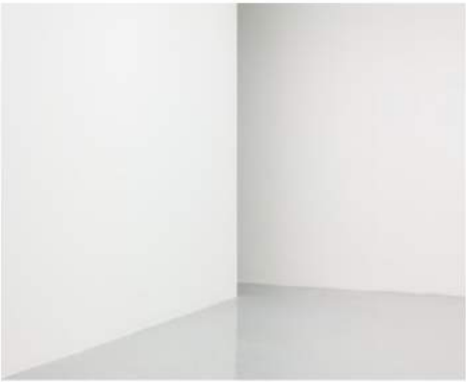
Recap: multi-view stereo



Cost volume is a 3D array that stores the errors of all pixels at all depths

Learning MVS

Challenges for traditional methods



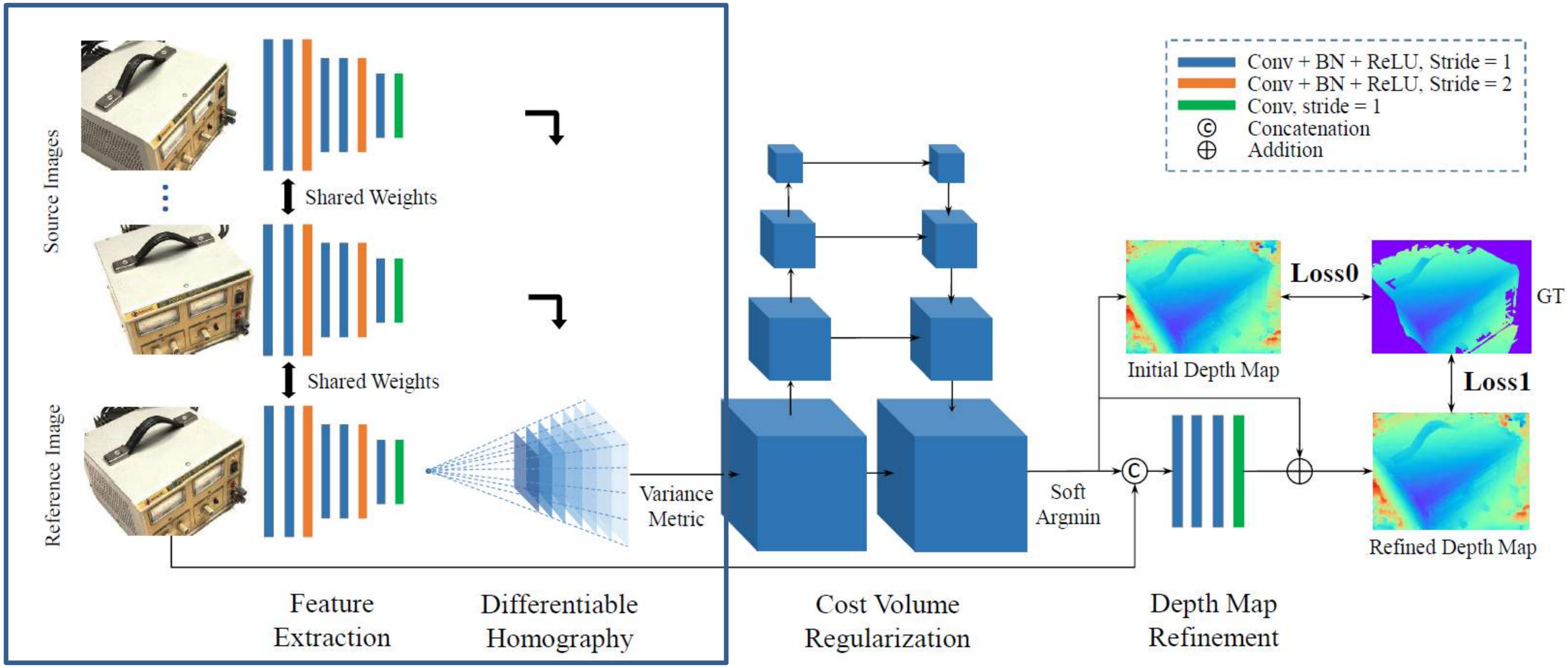
Textureless Area

Reflection
/Transparency

Repetitive
patterns

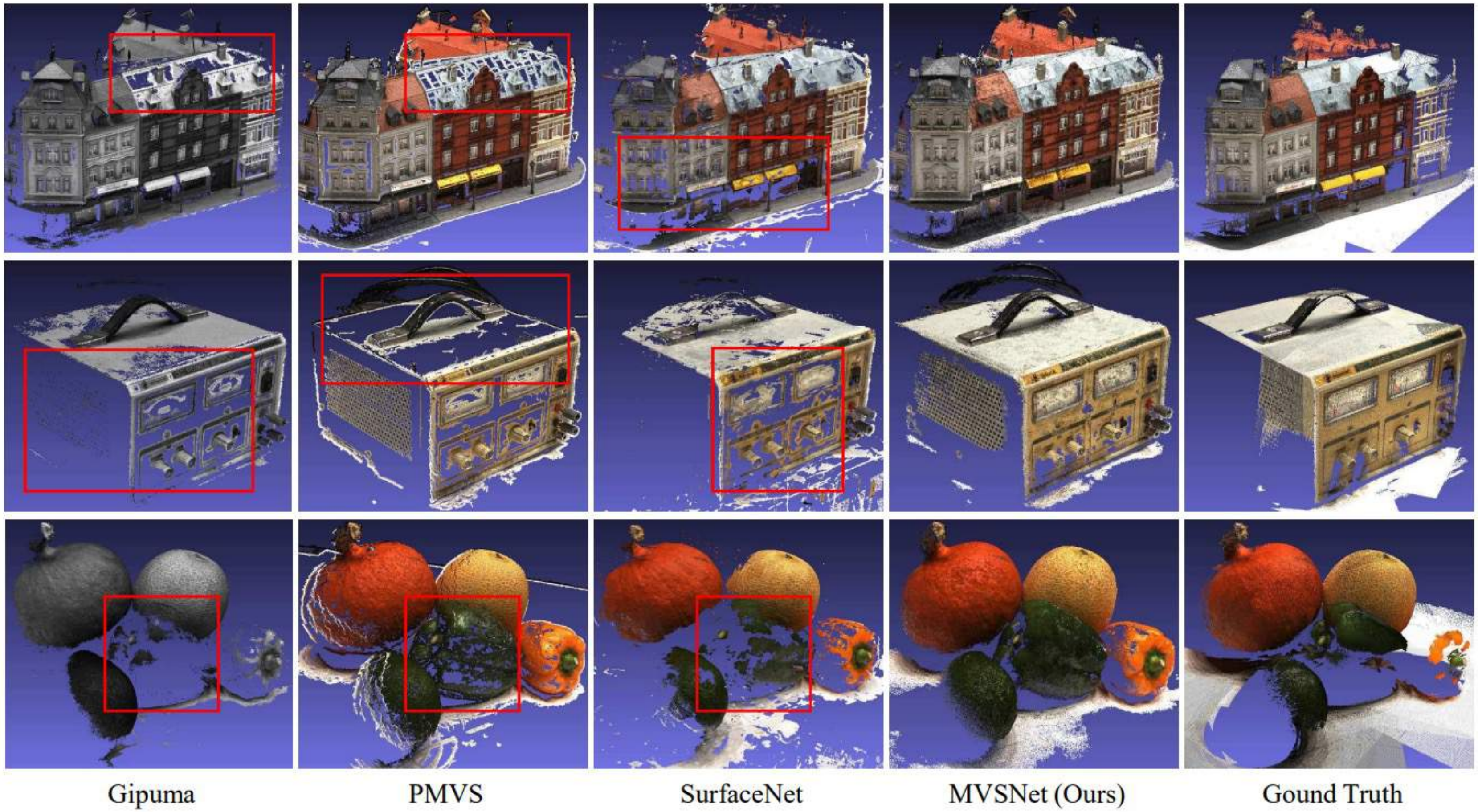
Learning MVS

MVSNet: predict cost volume from CNN features



Yao, Yao, et al. "Mvsnet: Depth inference for unstructured multi-view stereo." ECCV 2018

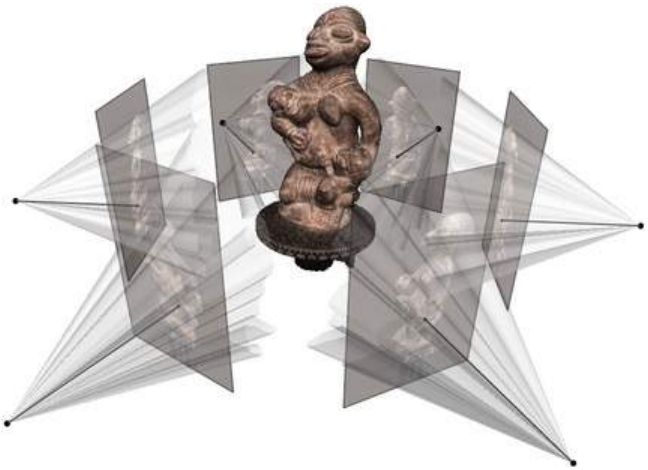
Learning MVS



Yao, Yao, et al. “Mvsnet: Depth inference for unstructured multi-view stereo.” ECCV 2018

Dense Reconstruction

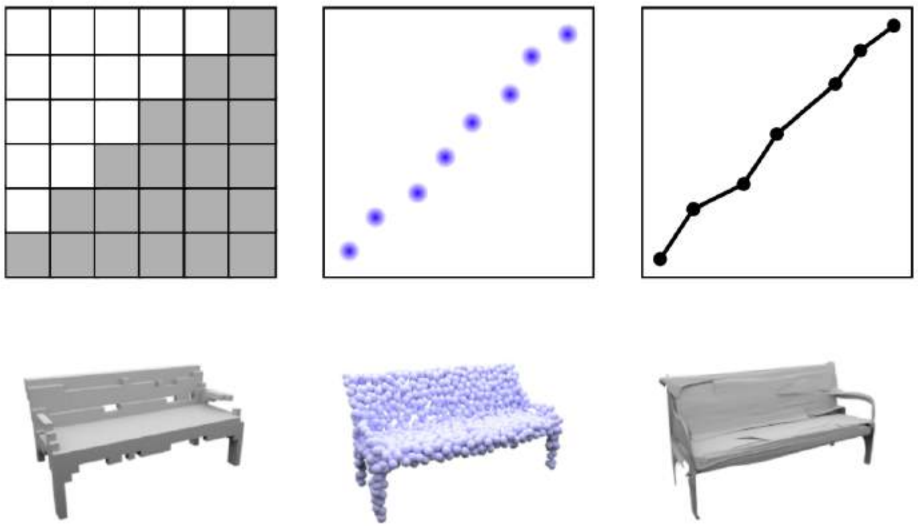
Can we improve the mesh quality by comparing the rendered images with the input images?



Yes, but not easy

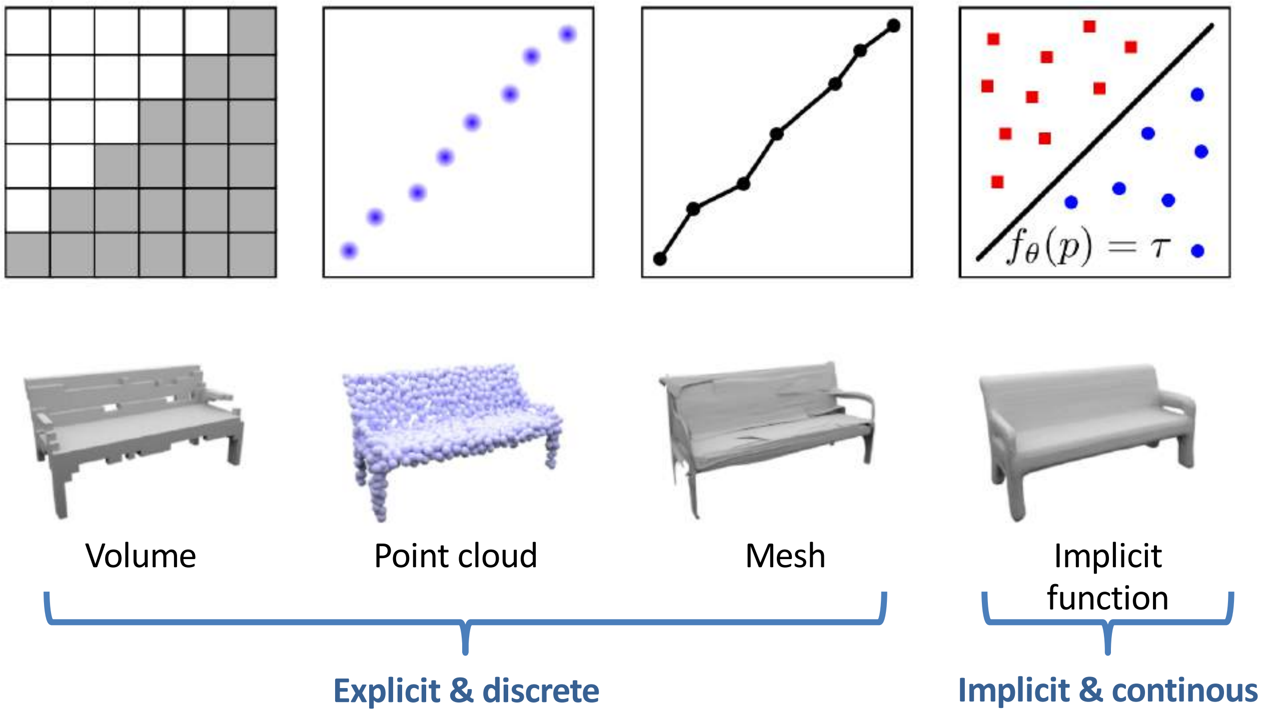
- The rendering process is not differentiable
- Mesh representation is not a good representation for optimization

3D Representations



Volume | Point cloud | Mesh

Implicit Representations



Implicit Representations

A simple example:

$f(p) = 1$ represents a sphere

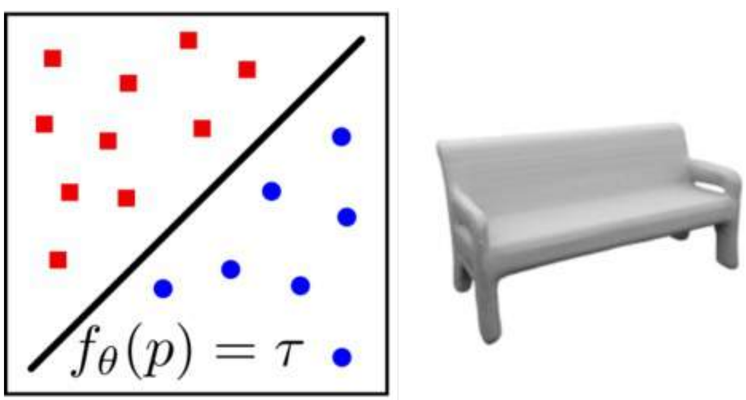
where

$f(p) = p_1^2 + p_2^2 + p_3^2$

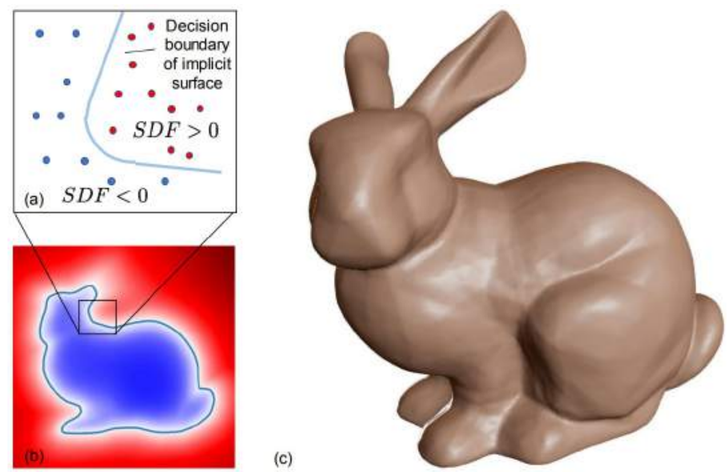
Implicit Representations

Generally, the implicit function can be:

Occpupacy



Signed distance function (SDF)



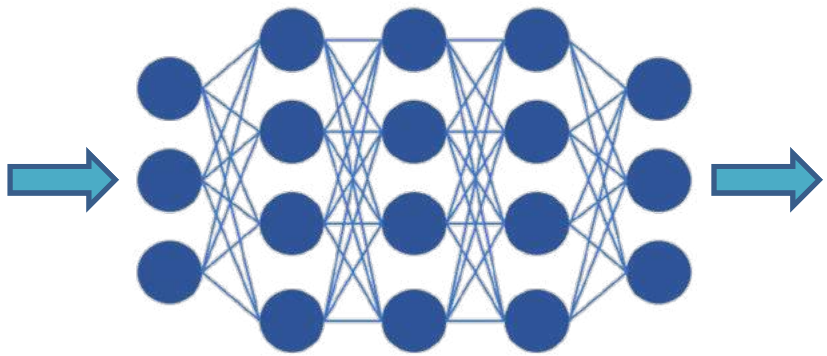
What is the form of $f_\theta(p)$?

- In practice, very hard to define!

Implicit Neural Representations

MLP

3D coordinates



Occupancy
Signed Distance
…

Occupancy Networks: Learning 3D Reconstruction in Function Space, *CVPR* 2019.

DeepSDF: Learning Continuous Signed Distance Functions for Shape Representation, *CVPR* 2019.

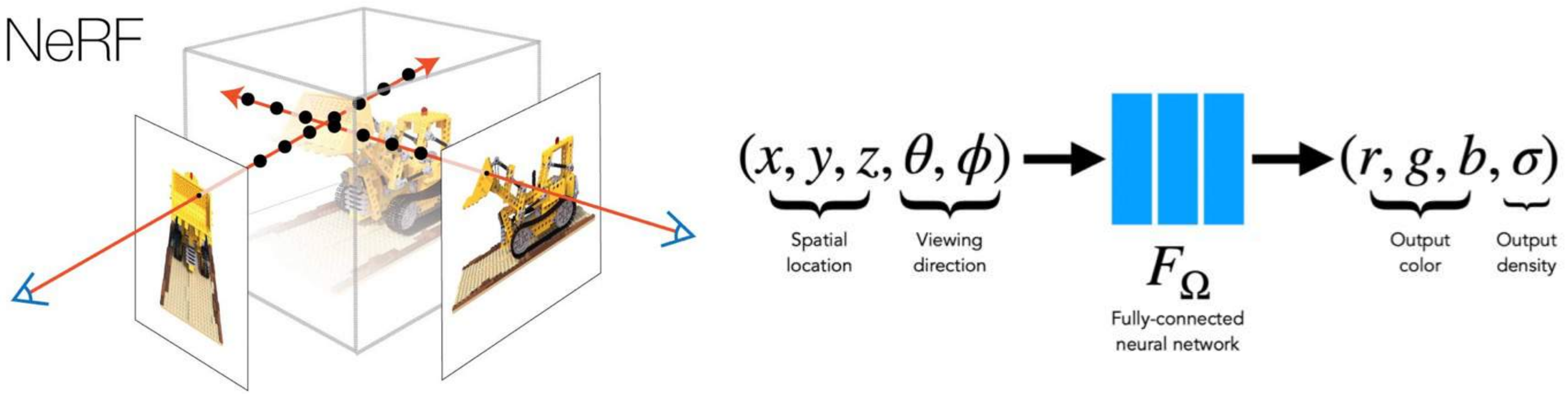
Learning implicit fields for generative shape modeling, *CVPR* 2019.

Scene Representation Networks: Continuous 3D-Structure-Aware Neural Scene Representations, NeurIPS 2019.

Neural Radiance Fields (NeRF)

Representing scenes as radiance fields

- Radiance = density + color



NeRF: Representing scenes as neural radiance fields for view synthesis, *ECCV* 2020.

Neural Radiance Fields (NeRF)

The radiance filed can be converted to images by volume rendering, which is differentiable

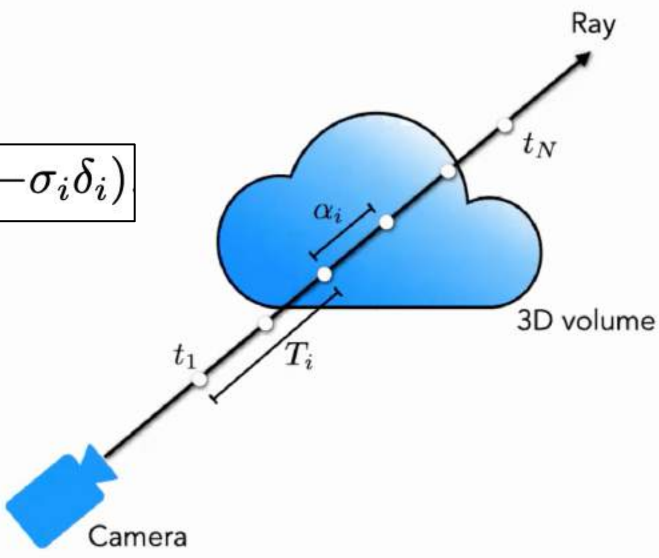
Rendering model for ray r(t) = o + td:

$$C \approx \sum_{i=1}^{N} \underbrace{T_i}_{\text{weights}} \alpha_i \underbrace{c_i}_{\text{colors}}$$

$$\boxed{\alpha_i = 1 - \exp(-\sigma_i \delta_i)}$$

How much light is blocked earlier along ray:

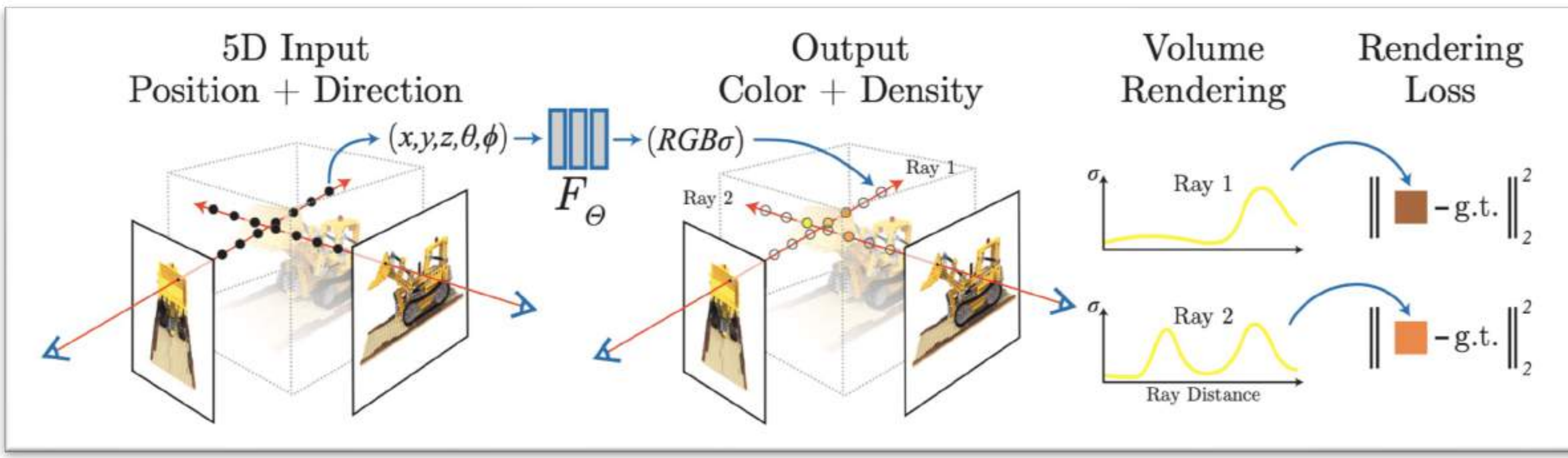
$$T_i = \prod_{j=1}^{i-1} (1 - \alpha_j)$$



NeRF: Representing scenes as neural radiance fields for view synthesis, *ECCV* 2020.

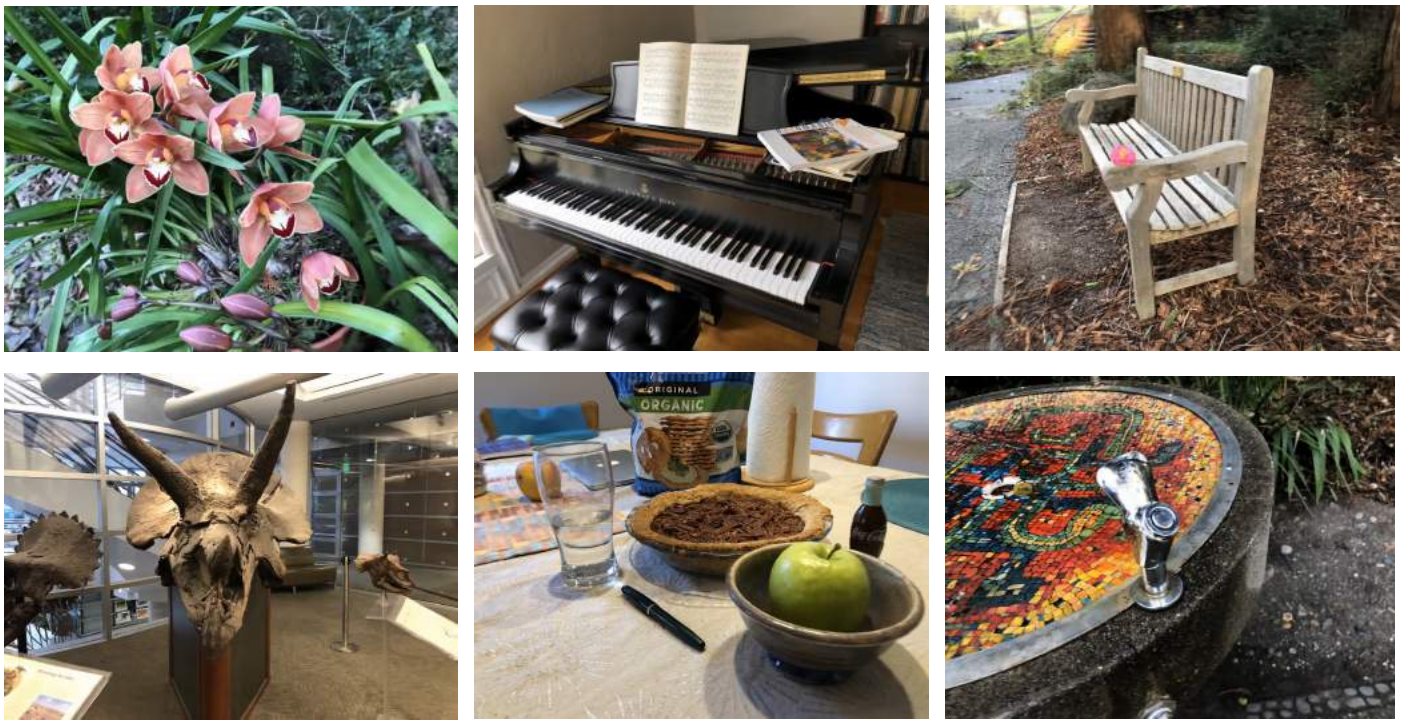
Neural Radiance Fields (NeRF)

Reconstructing NeRF from images by minimizing rendering loss



NeRF: Representing scenes as neural radiance fields for view synthesis, *ECCV* 2020.

Neural Radiance Fields (NeRF)

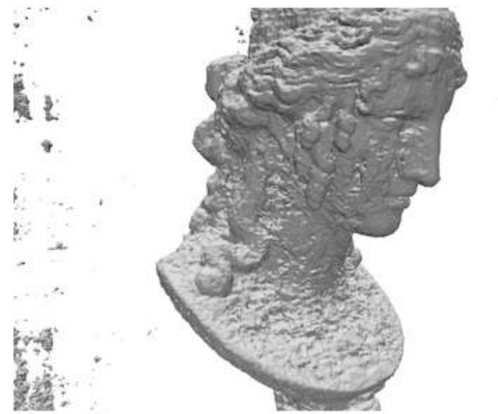


Neural Radiance Fields (NeRF)

But NeRF has poor surface reconstruction quality

Reference image

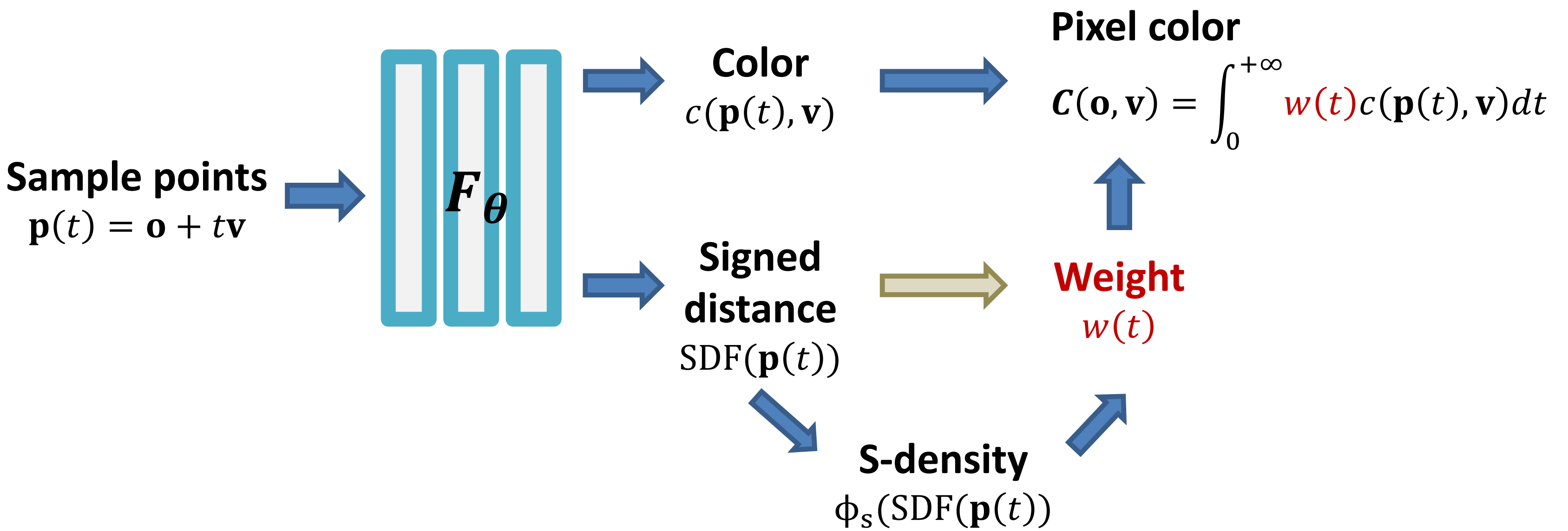
COLMAP



NeRF

NeuS

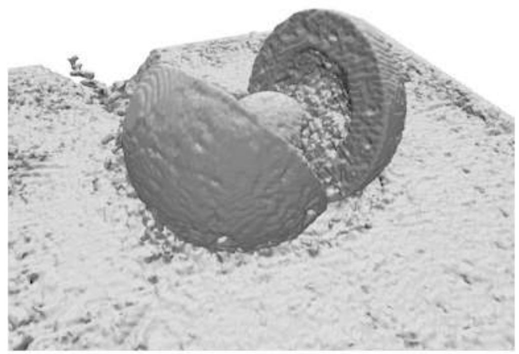
Replacing density field by SDF



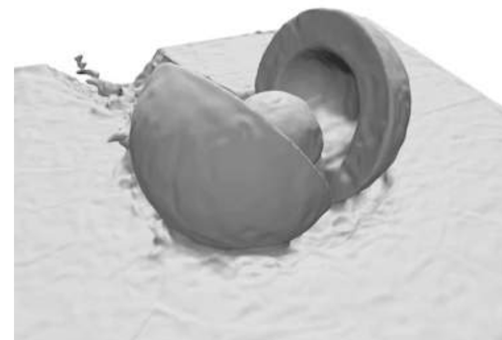
NeuS: Learning Neural Implicit Surfaces by Volume Rendering for Multi-view Reconstruction, *NeurIPS* 2021.

NeuS

Reference image



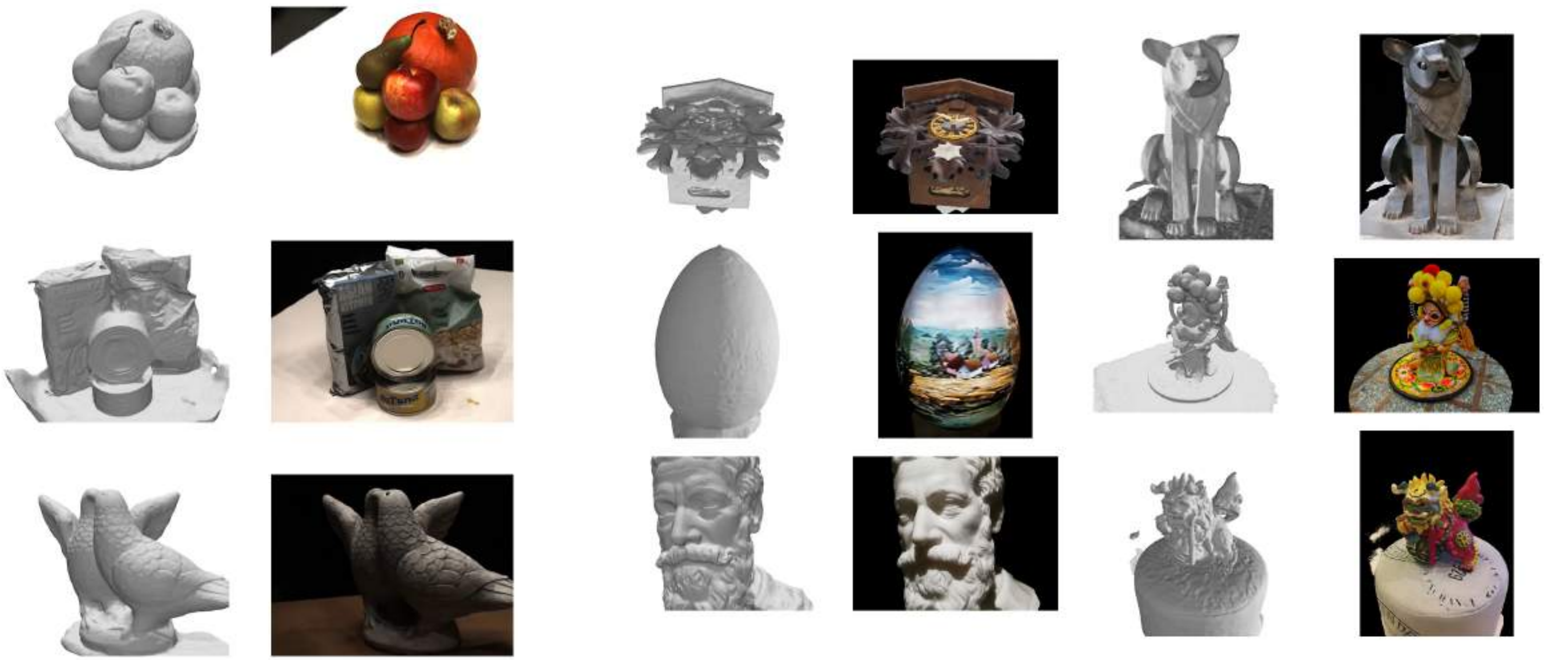
NeRF



NeuS

NeuS: Learning Neural Implicit Surfaces by Volume Rendering for Multi-view Reconstruction, *NeurIPS* 2021.

NeuS

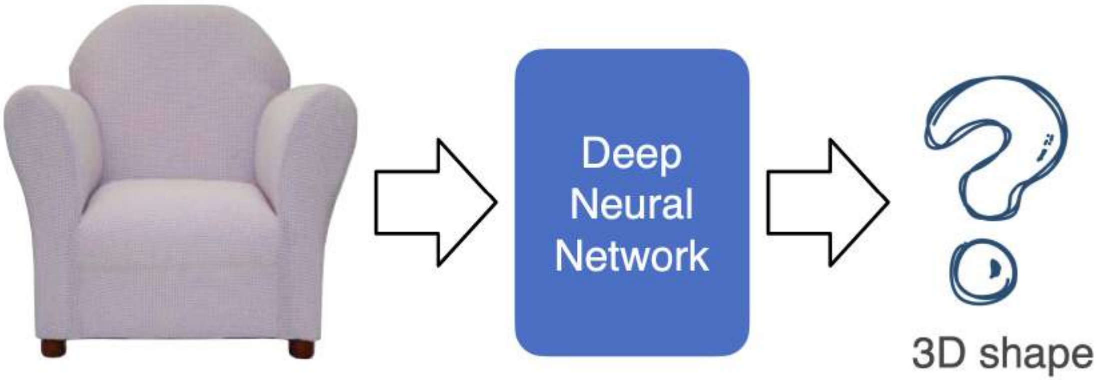


NeuS: Learning Neural Implicit Surfaces by Volume Rendering for Multi-view Reconstruction, *NeurIPS* 2021.

Single image to 3D

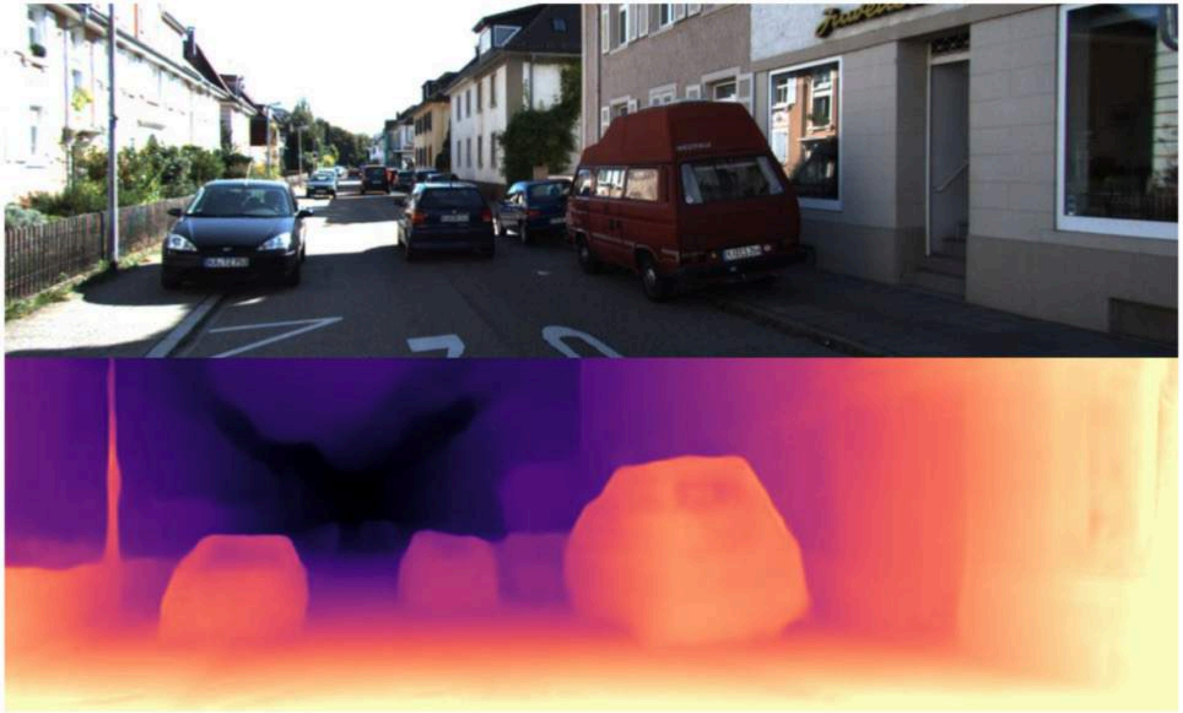
Infer 3D representation from just single image

- Depth
- Point Cloud
- Mesh
- Volume



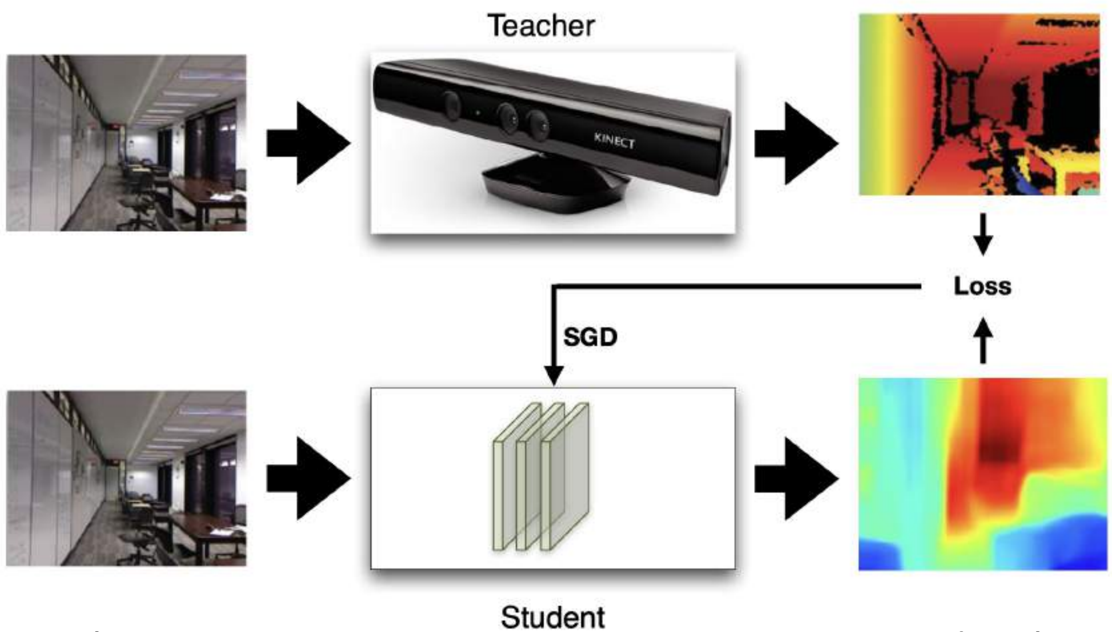
Monoculer depth estimation

Using network to guess depth from single image



Monoculer depth estimation

Using network to guess depth from single image

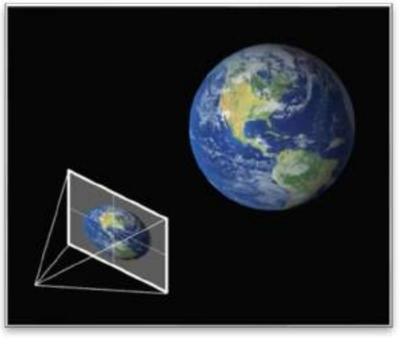


Single Image

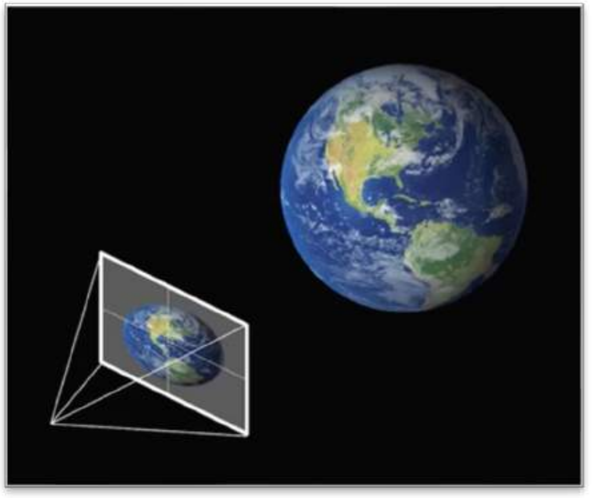
Output depth map

Monoculer depth estimation

Scale ambiguity: the same object with different sizes and depths give the same image



Scale = k_1



Scale = k_2

Monoculer depth estimation

Loss function: scale-invariant depth error

Standard L2 error

$$D_{L2}(y, y^*) = \frac{1}{n} \sum_{i=1}^{n} (\log y_i - \log y_i^*)^2$$

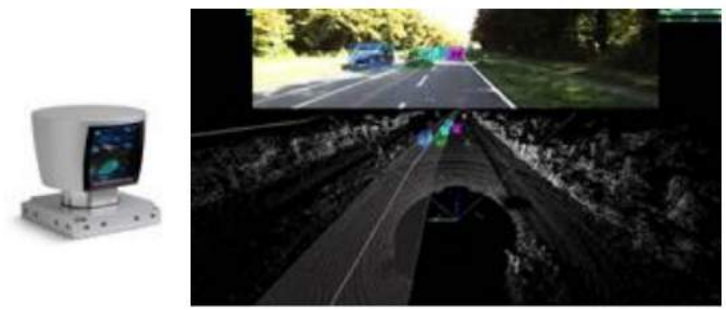
Scale invariant error

$$D_{SI}(y, y^*) = \frac{1}{n} \sum_{i=1}^{n} (\log y_i - \log y_i^* + \alpha(y, y^*))^2$$

with $\alpha(y, y^*) = \frac{1}{n} \sum_{j=1}^{n} (\log y_j - \log y_j^*)$

Monoculer depth estimation

Where is training data from



KITTI [Geiger et al. 2012]

NYU [Eigen et al. 2014]

Depth in the Wild [Chen et al. 2016]

Direct, real-world training data is limited for geometric problems

Monoculer depth estimation

MegaDepth dataset

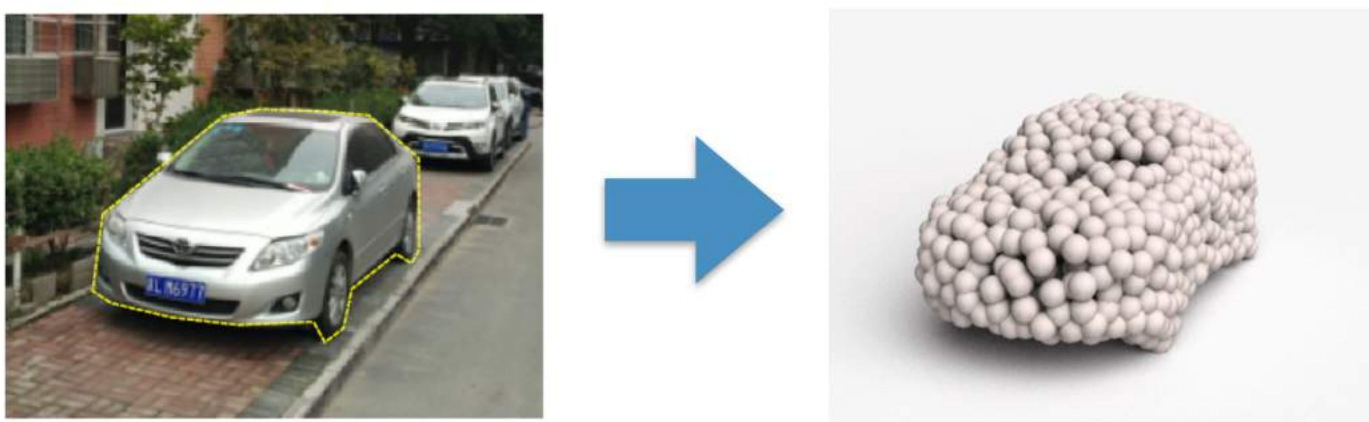
>130K (RGB, depth map) pairs

- Ultilize images on the Internet!
- Generated from 200+ landmarks in the world
- Reconstructed with SfM + MVS using COLMAP [Schoenberger et al]

Zhengqi Li and Noah Snavely. MegaDepth: Learning Single-View Depth Prediction from Internet Photos. CVPR 2018

Single-view shape estimation

3D model from a single image

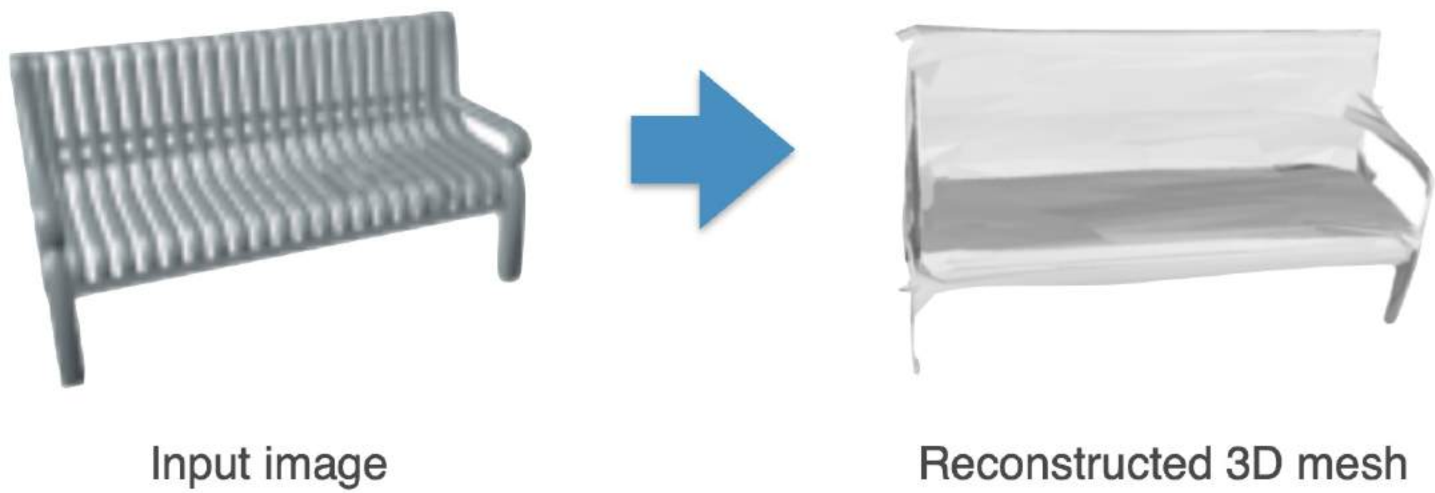


Input image Reconstructed 3D point cloud

Fan et al., “A Point Set Generation Network for 3D Object Reconstruction from a Single Image”, CVPR 2017

Single-view shape estimation

3D model from a single image

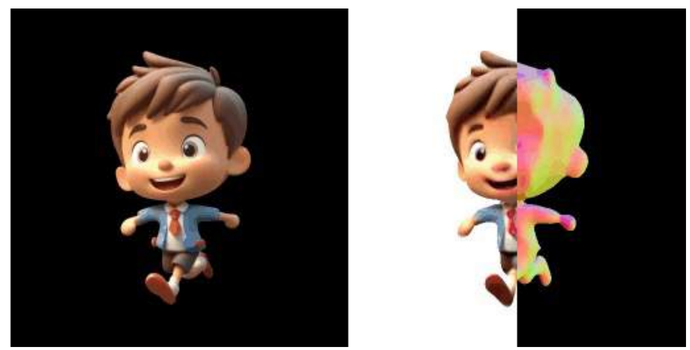


Groueix et al, AtlasNet:A papier-mâché approach to
learning 3d surface generation, CVPR 2018

3D generation nowadays

ProlificDreamer DreamFusion Magic3D SJC LatentNeRF Fantasia3D TextMesh

Zero-1–to-3



Magic123

https://github.com/threestudio-project/threestudio

Questions?

Today

Deep Learning for 3D reconstruction

1. Feature matching
2. Object Pose Estimation
3. Human Pose Estimation
4. Dense Reconstruction

Deep learning for 3D understanding

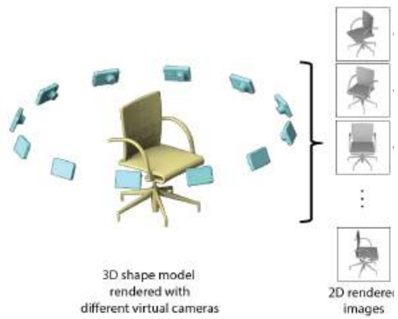
3D Classification

Input: 3D shape

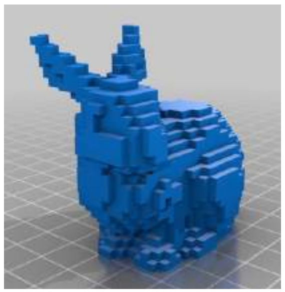
Output: Class

This is a chair!

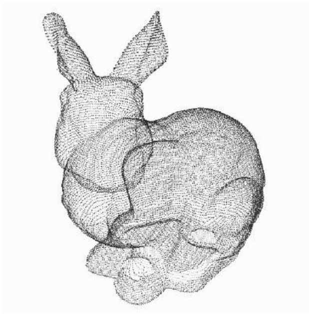
3D Data Representations



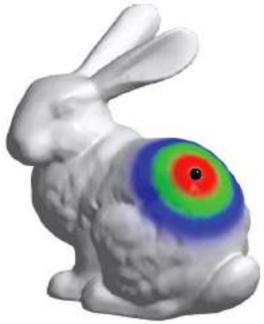
Multi-view



Volumetric



Point Cloud



Mesh

$$F(x) = 0$$

Implicit Shape

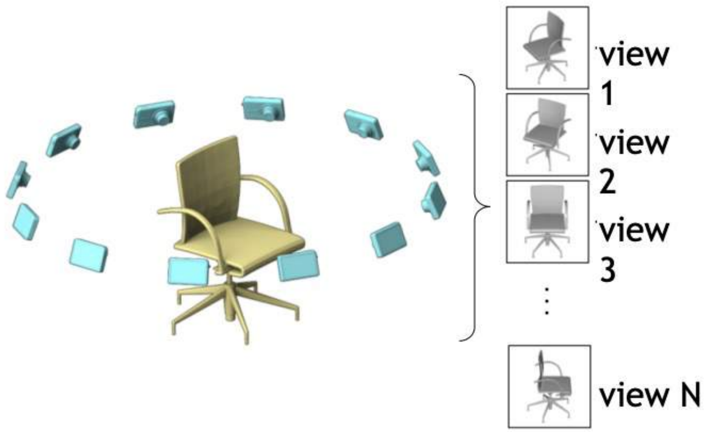
Multi-view CNNs

1. Given an input shape

Su et al., "Multi-view Convolutional Neural Networks for 3D Shape Recognition”, ICCV 2015

Multi-view CNNs

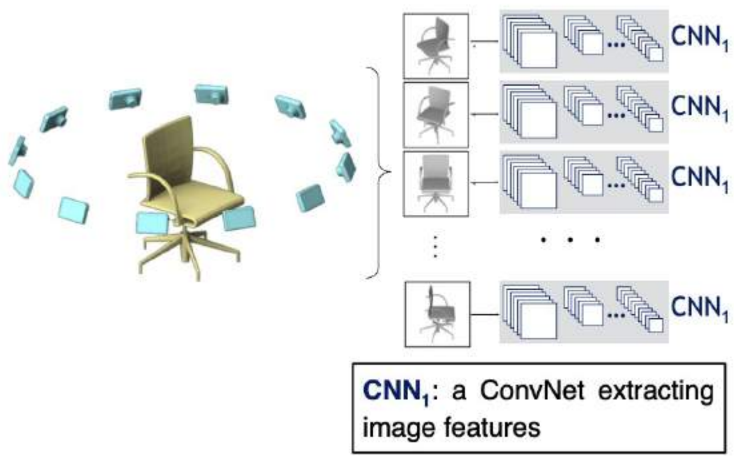
2. Render with multiple virtual cameras



Su et al., "Multi-view Convolutional Neural Networks for 3D Shape Recognition", ICCV 2015

Multi-view CNNs

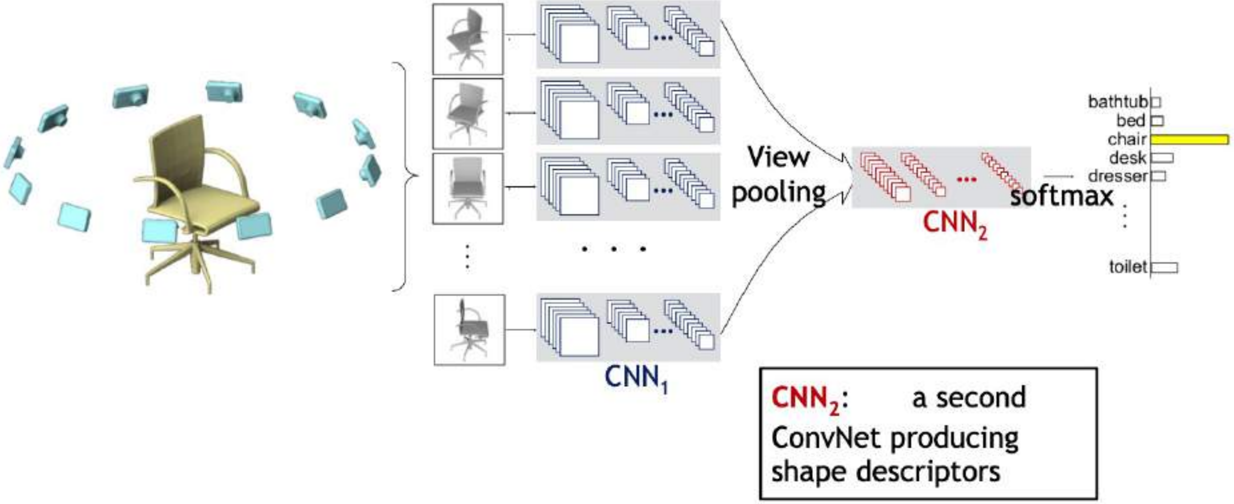
3. The rendered images are passed through CNNs



Su et al., "Multi-view Convolutional Neural Networks for 3D Shape Recognition", ICCV 2015

Multi-view CNNs

3. All image features are pooled and then passed through CNNs and to generate final predictions



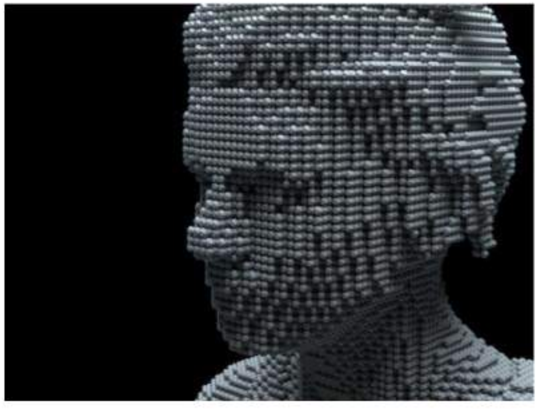
Su et al., "Multi-view Convolutional Neural Networks for 3D Shape Recognition", ICCV 2015

Deep learning on volumetric data

How to process 3D volumetric data?

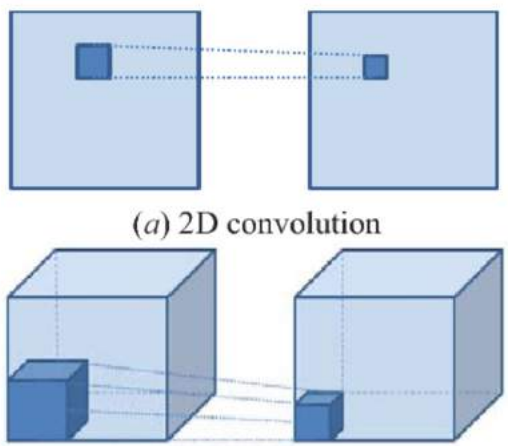
Voxel: occupancy

$$V_{ijk} = \begin{cases} 1 & \text{if occupied} \\ 0 & \text{if empty} \end{cases}$$



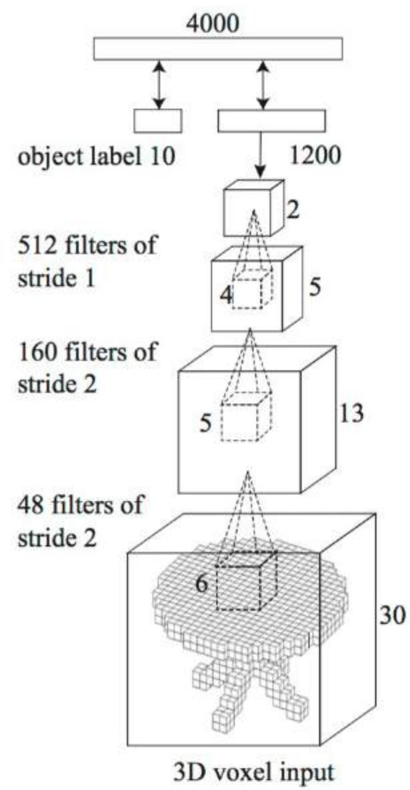
3D ConvNets

Process volumetric data using 3D convlution



(a) 2D convolution

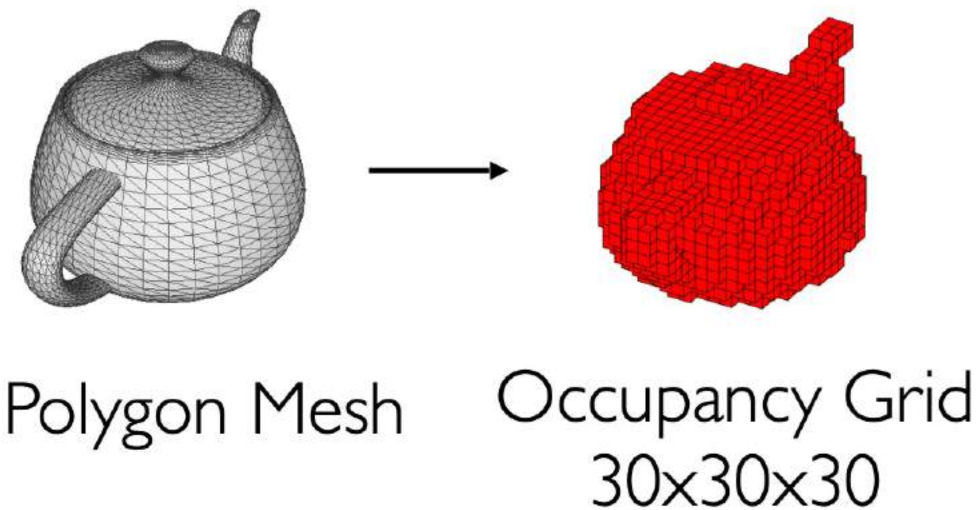
(b) 3D convolution



3DShapeNets
(Wu et al, 2015)

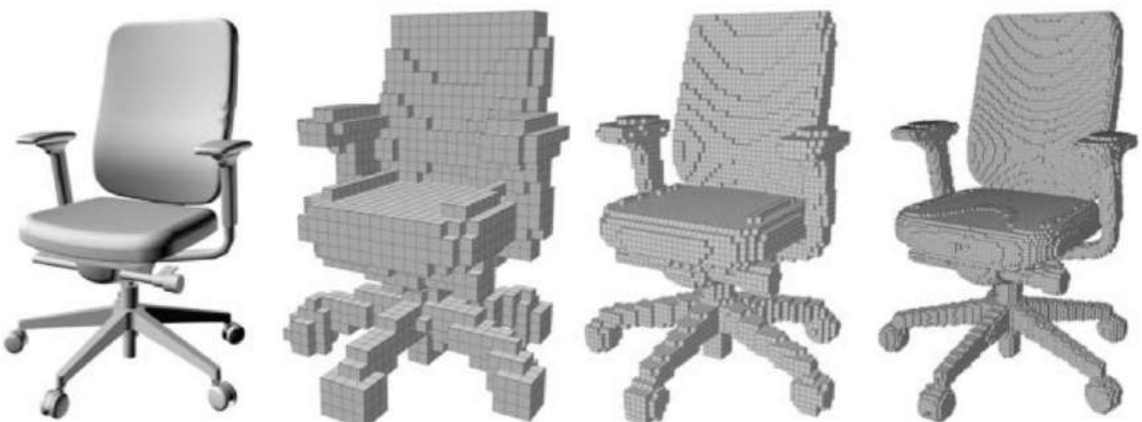
3D ConvNets

Challenge: High space/time complexity of high resolution voxels: $O(N^3)$



Sparse ConvNets

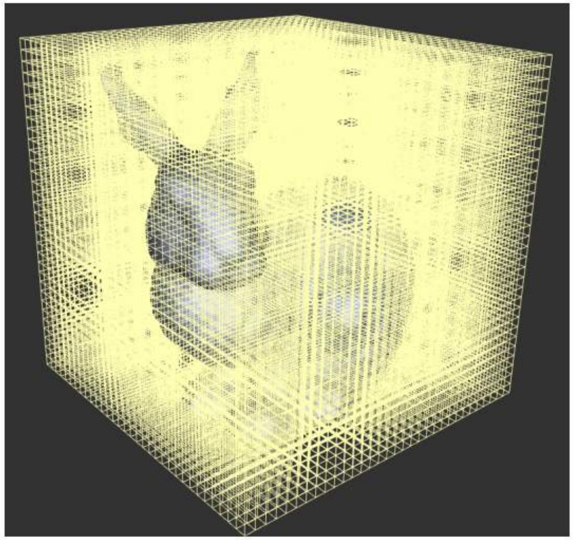
Idea: using sparity of 3D shapes

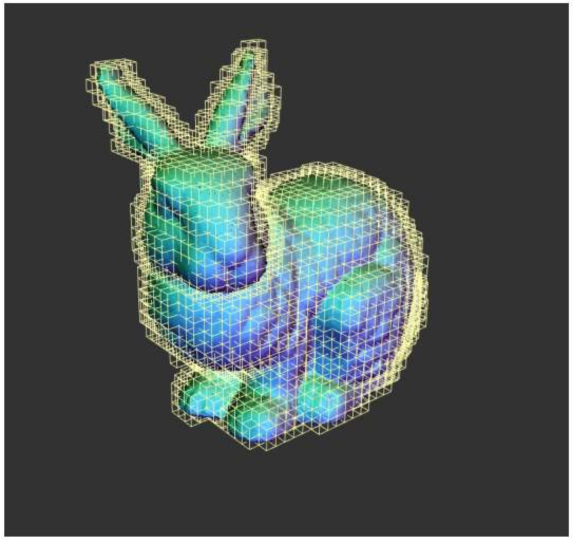


$\frac{\#occupied\ grid}{\#total\ grid}$	10.41%	5.09%	2.41%
Resolution:	32	64	128

Sparse ConvNets

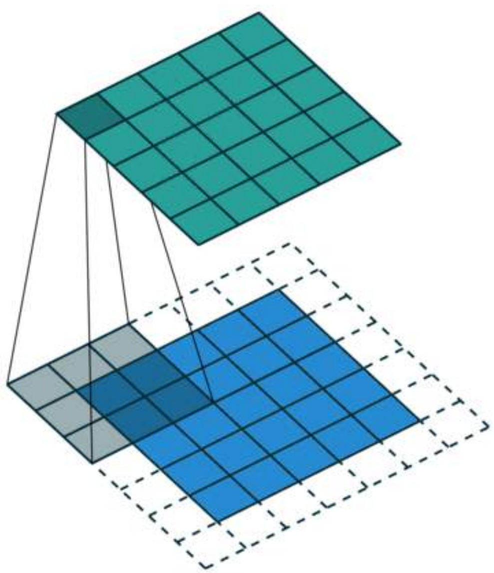
- Store the sparse surface signals (Octree)
- Constrain the computation near the surface



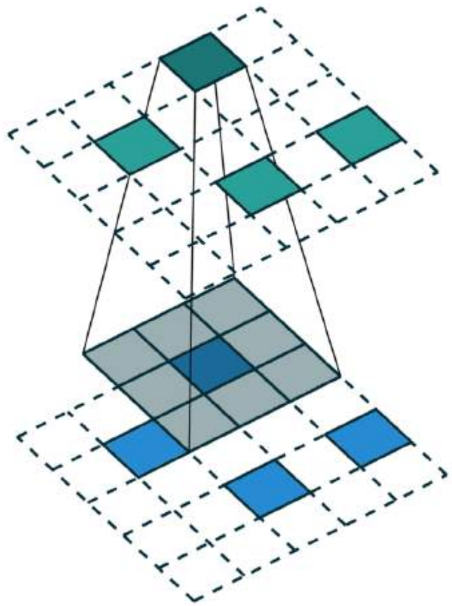


Sparse ConvNets

Sparse convolution: compute inner product only at the active sites (nonzero entries)



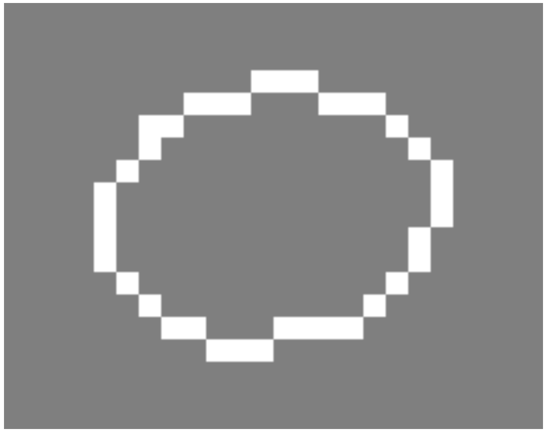
Regular convolution



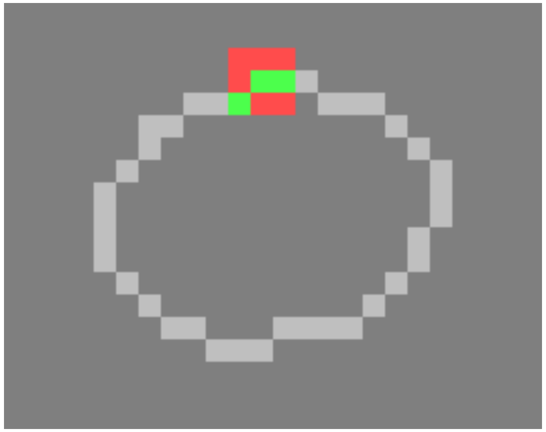
Sparse convolution

Sparse ConvNets

Sparse convolution: compute inner product only at the active sites (nonzero entries)



With regular 3x3 convolutions, the set of active (non-zero) sites grows rapidly

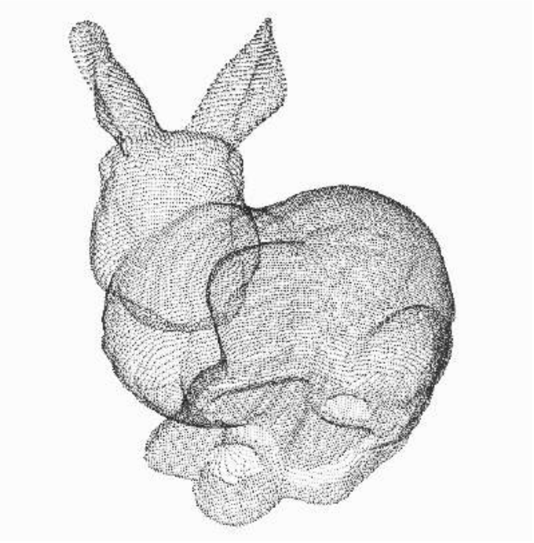


With **Sparse Convolutions**, the set of active sites is unchanged

https://github.com/facebookresearch/SparseConvNet

Deep learning on point clouds

How to process point cloud using neural networks?



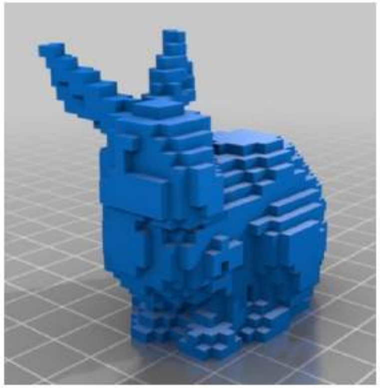
Point cloud

(The most common 3D sensor data)

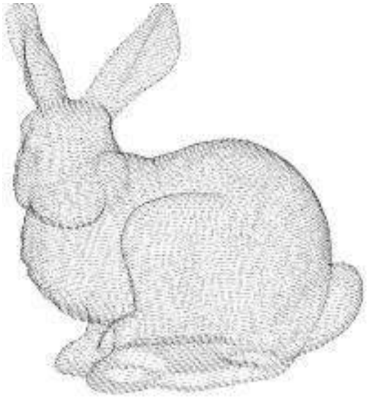
Deep learning on point clouds

Challenges:

- Point cloud is unrasterized data
- Convolution cannot be applied



VS

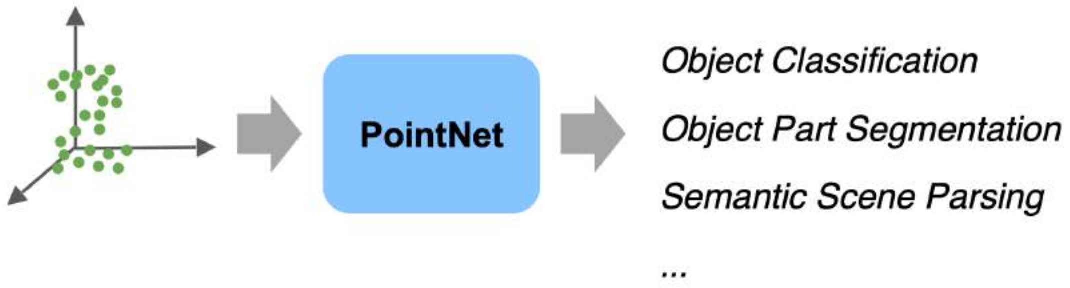


Rasterized form
(regular grid)

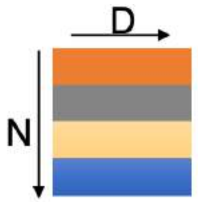
Unrasterized
(Irregular)

PointNet

A point cloud processing architecture for multiple tasks
(classification, detection, segmentation, registration, etc)



Point cloud: N orderless points, each represented by a D dim coordinate



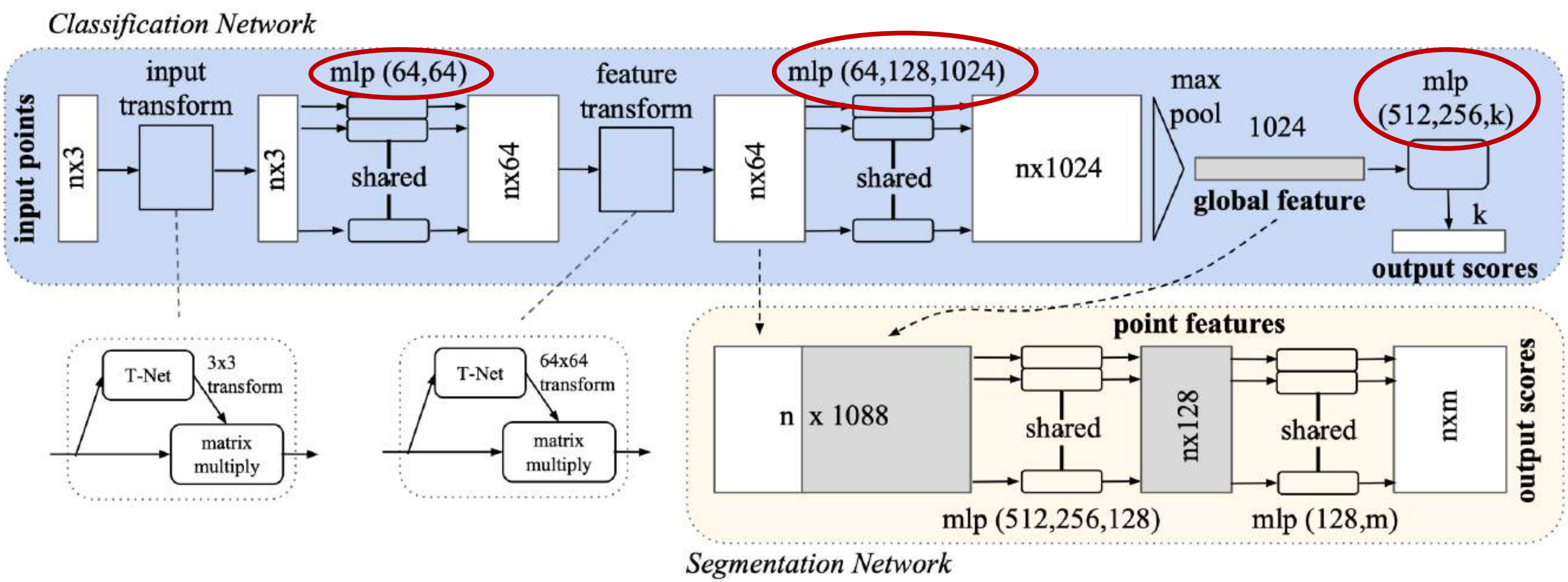
2D array representation

Qi, Charles R., et al. "Pointnet: Deep learning on point sets for 3d classification and segmentation." CVPR 2017

PointNet

PointNet Classification and Segmentation Architecture

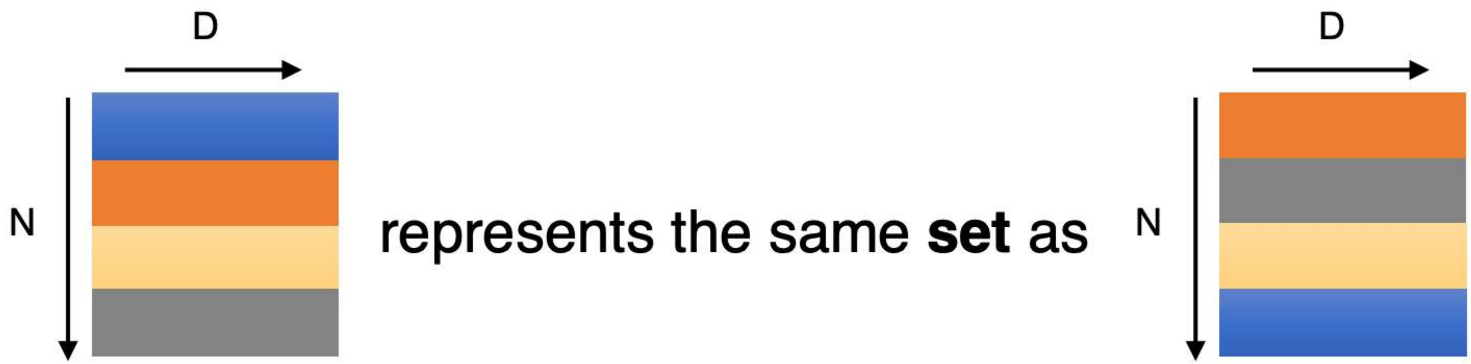
mlp: multi-layer perceptron (fully connected)



Qi, Charles R., et al. "Pointnet: Deep learning on point sets for 3d classification and segmentation." CVPR 2017

PointNet

Challenge 1: the point set is order-less

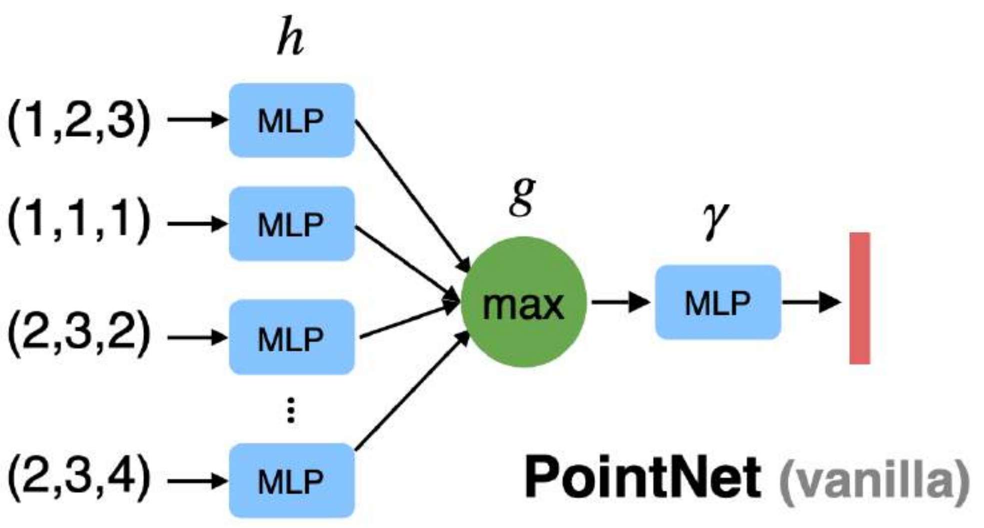


The output needs to be invariant to N! permutations

Qi, Charles R., et al. "Pointnet: Deep learning on point sets for 3d classification and segmentation." CVPR 2017

PointNet

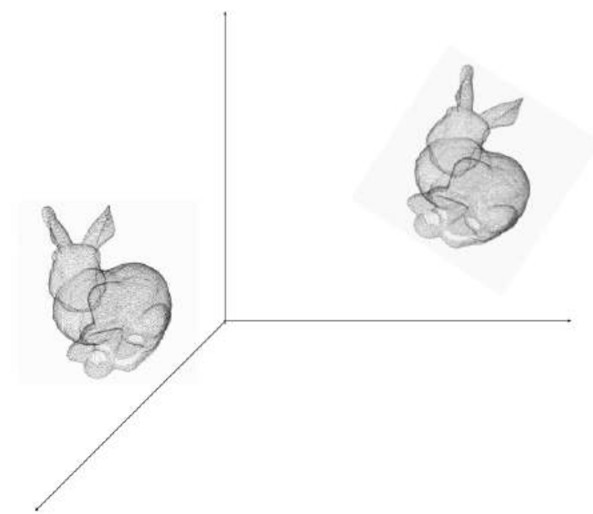
Solution: max pooling makes the output invariant to the order of input points



Qi, Charles R., et al. "Pointnet: Deep learning on point sets for 3d classification and segmentation." CVPR 2017

PointNet

Challenge 2: the output should be invariant to rigid transformation of points

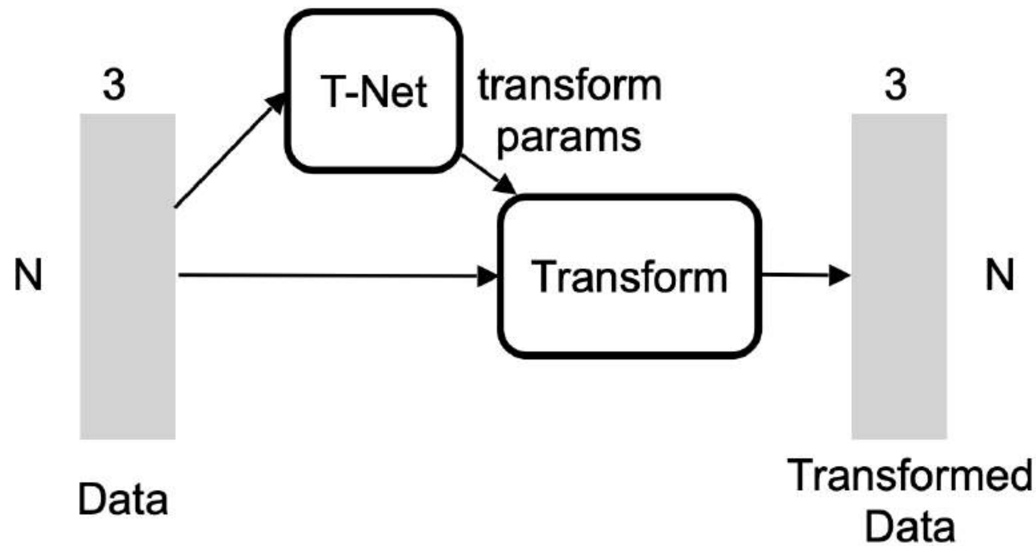


Let S be a shape. Then $f(T \cdot S) = f(S)$
f: classifier, T: transformation matrix

Qi, Charles R., et al. "Pointnet: Deep learning on point sets for 3d classification and segmentation." CVPR 2017

PointNet

Solution: estimate the transformation using another network (T-Net)



Qi, Charles R., et al. "Pointnet: Deep learning on point sets for 3d classification and segmentation." CVPR 2017

PointNet

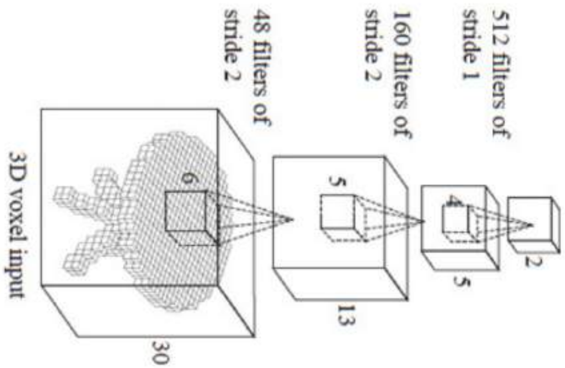
dataset: Stanford 2D-3D-S (Matterport scans)

PointNet++

Limitations of PointNet

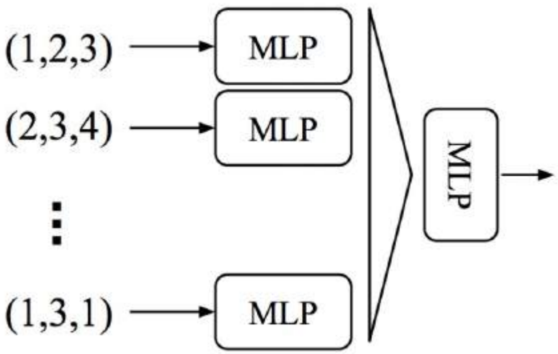
- No local context for each point!

Hierarchical feature learning
Multiple levels of abstraction



3D CNN (Wu et al.)

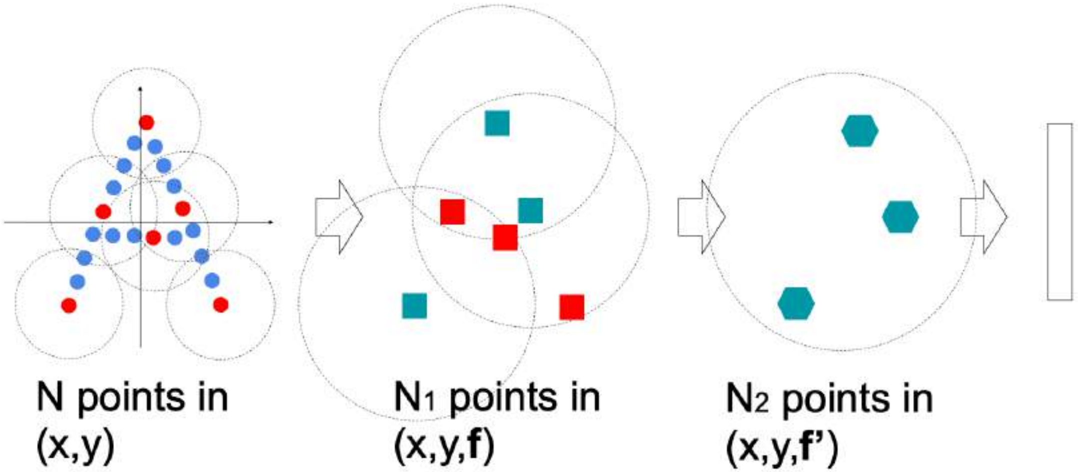
Global feature learning
Either one point or all points



PointNet (vanilla) (Qi et al.)

PointNet++

PointNet++ : Multi-Scale PointNet



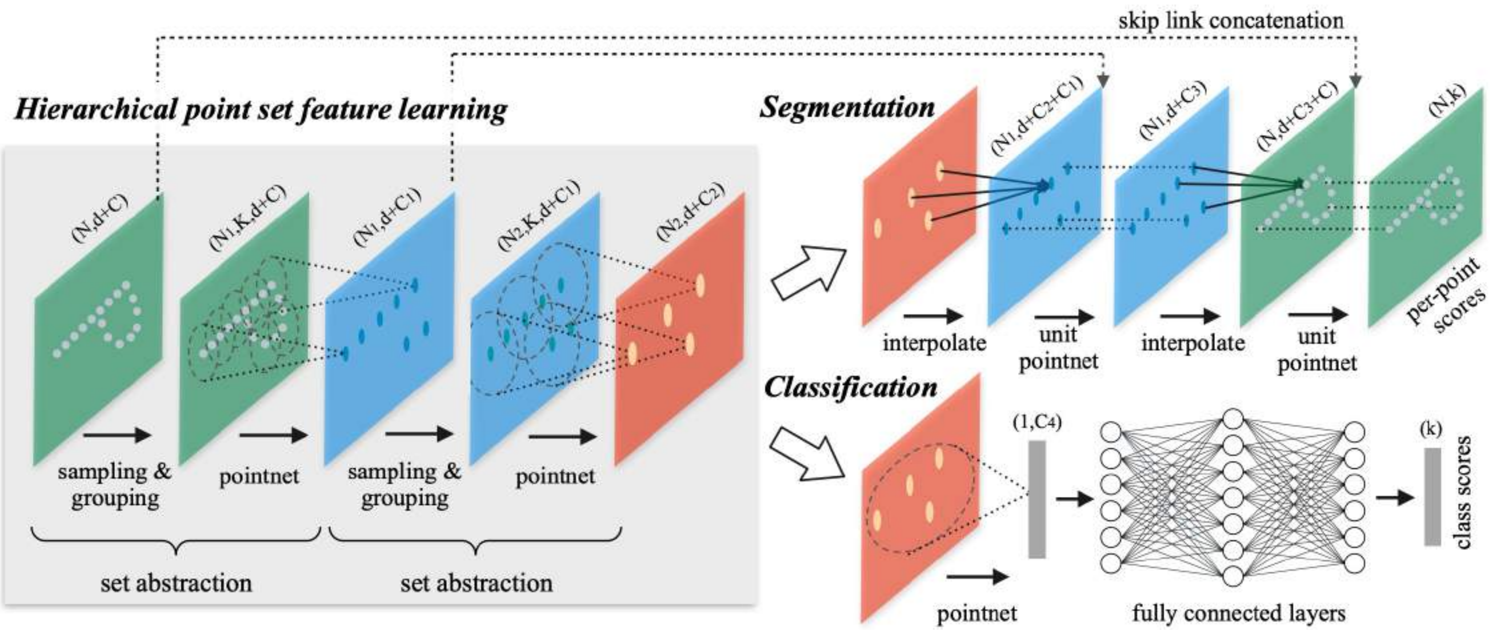
Three Parts:

- Sampling: Sample anchor points by Farthest Point Sampling (FPS)
- Grouping: Find neighbourhood of anchor points
- Apply PointNet in each neighborhood to mimic convolution

Qi, Charles R., et al. "Pointnet++: Deep hierarchical feature learning on point sets in a metric space."

PointNet++

PointNet++ : Multi-Scale PointNet



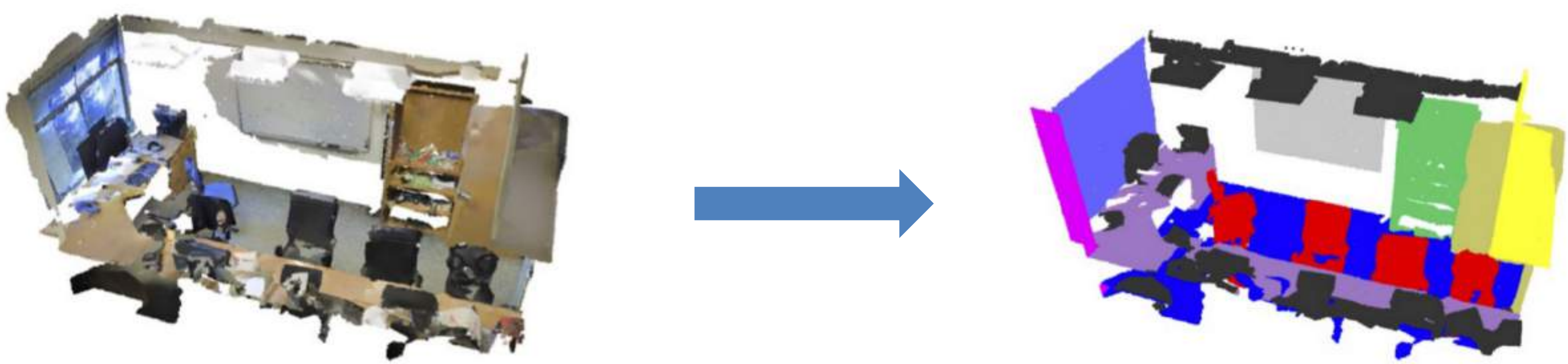
Qi, Charles R., et al. "Pointnet++: Deep hierarchical feature learning on point sets in a metric space."

3D semantic segmentation

Input: sensor data of a 3D scene (RGB/depth/point cloud .....)

Output: Label each point in point cloud with category label.

Possible solution: directly segmenting the point cloud

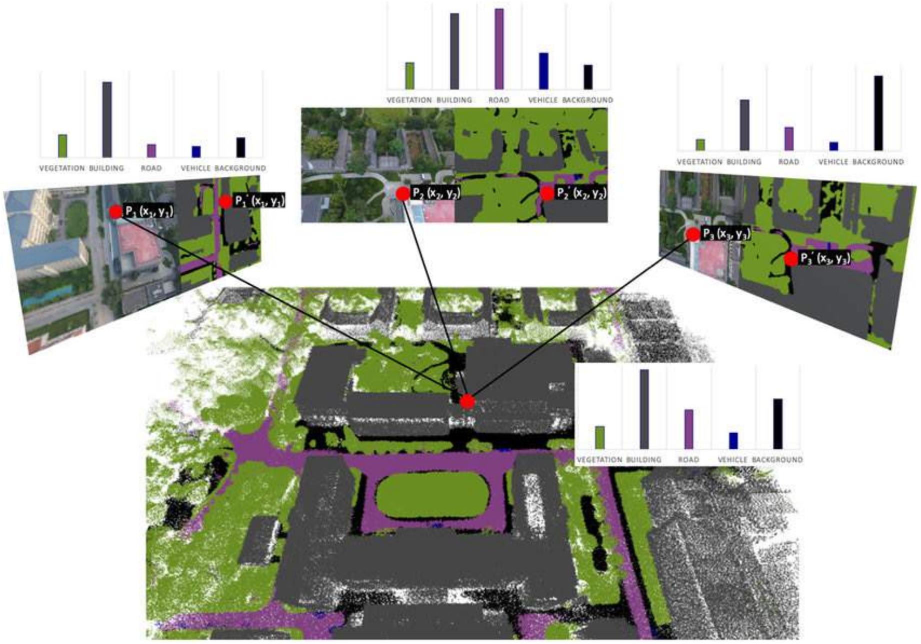


3D semantic segmentation

Input: sensor data of a 3D scene (RGB/depth/point cloud …..)

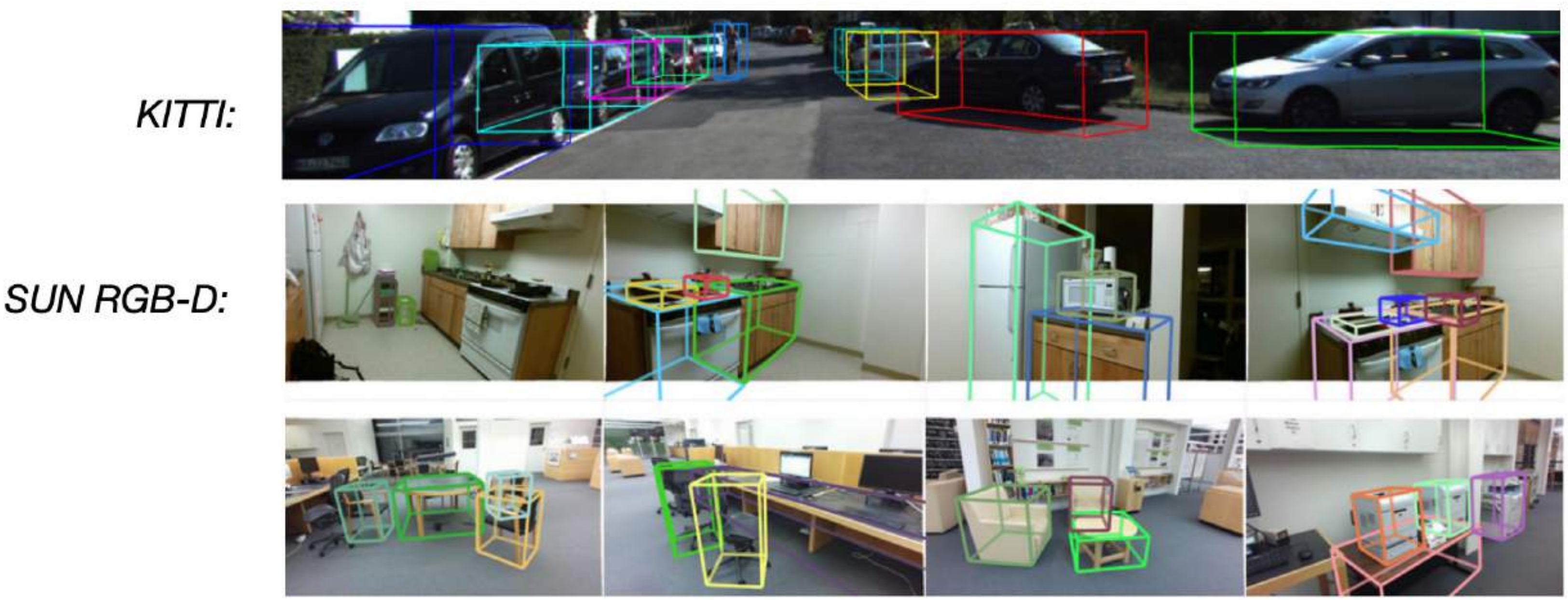
Output: Label each point in point cloud with category label.

Possible solution: fuse 2D segmentation results in 3D



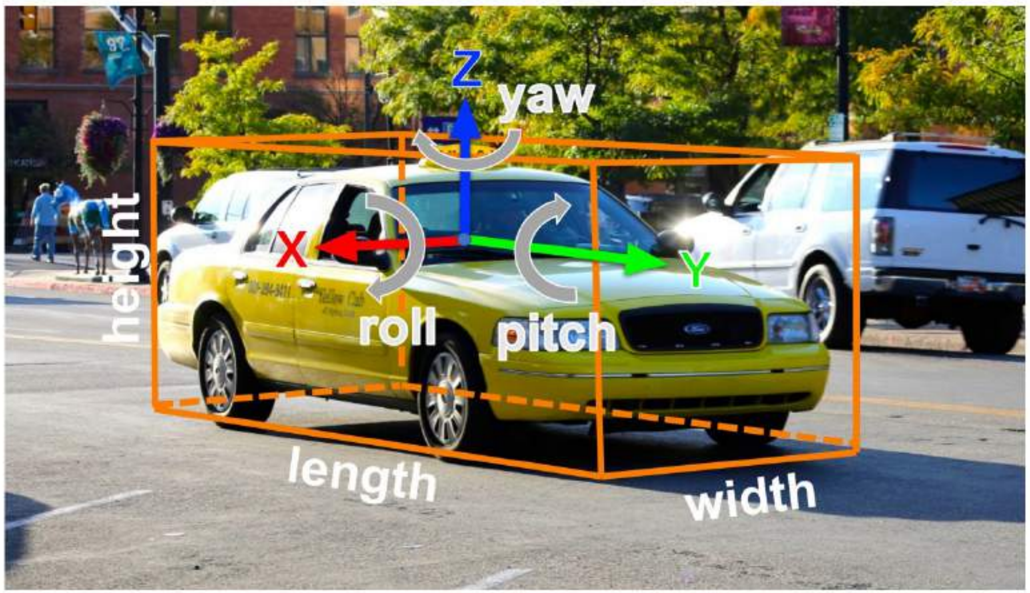
3D Object Detection

Detecting 3D objects in 3D data



3D Object Detection

3D bounding boxes



2D Object Detection:
2D bounding box
(x, y, w, h)

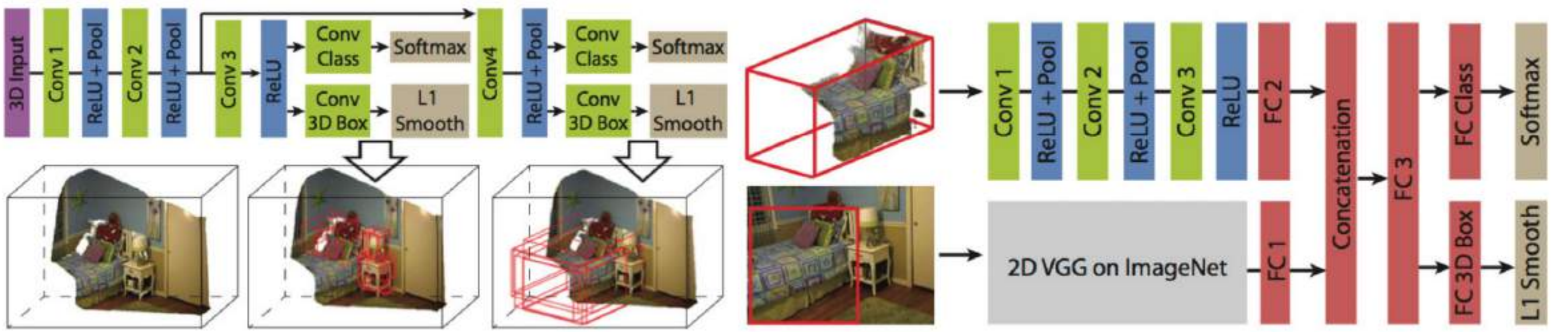
3D Object Detection:
3D oriented bounding box
(x, y, z, w, h, l, r, p, y)

Simplified bbox: no roll & pitch

Much harder problem than 2D object detection!

3D Object Detection

The first attempt: classify sliding windows

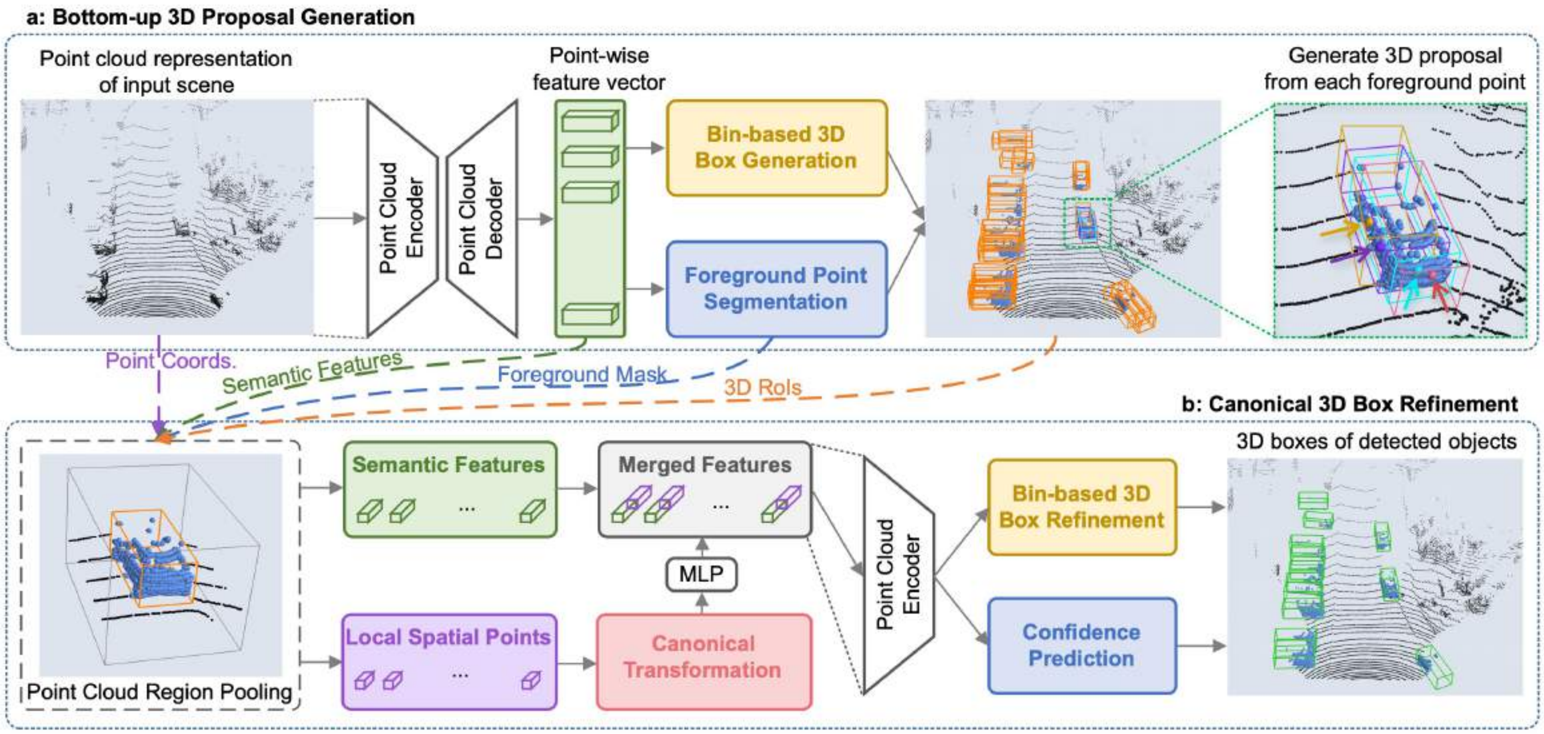


Con: 3D CNNs are very costly in both memory and time.

Song et al., "**Sliding Shapes for 3D Object Detection in Depth Images**", *ECCV 2014*

3D Object Detection

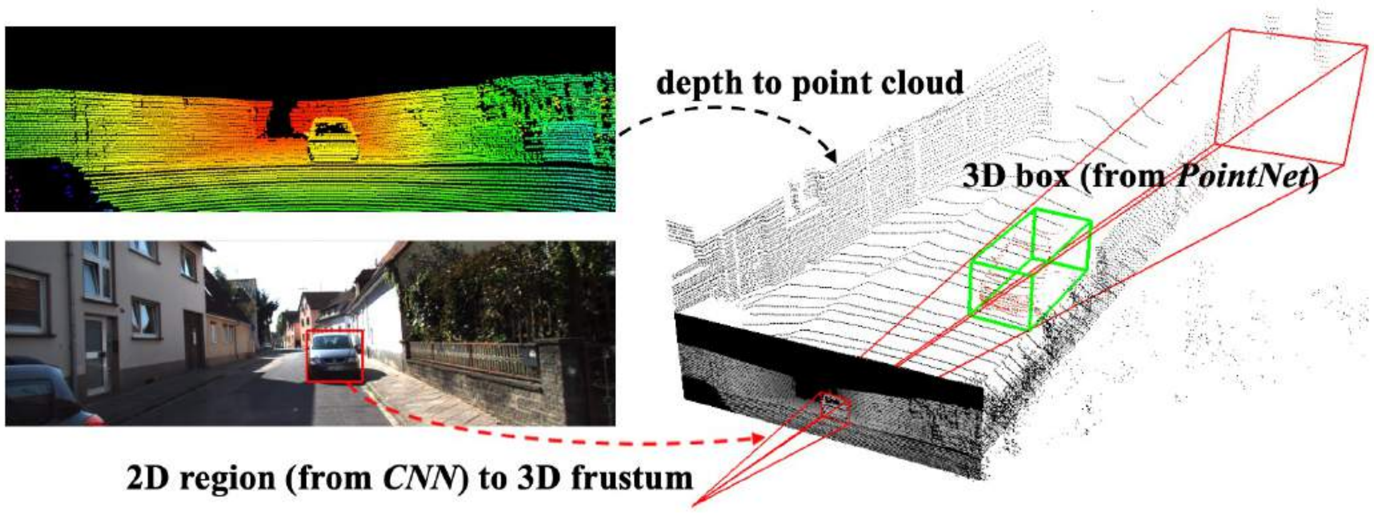
PointRCNN: RCNN for point cloud



Shi, Shaoshuai, *Xiaogang Wang*, and *Hongsheng Li*. "Pointrcnn: 3d object proposal generation and detection from point cloud." CVPR 2019

3D Object Detection

Frustum PointNets:
Using 2D detectors to generate 3D proposals

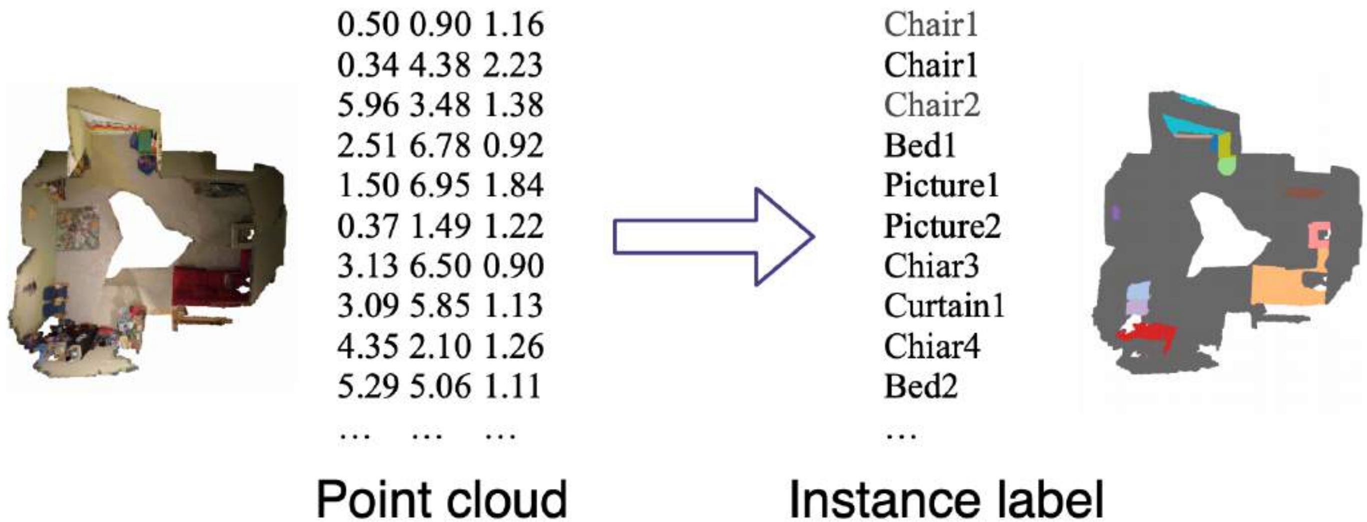


Qi, Charles R., et al. "Frustum pointnets for 3d object detection from rgb-d data." CVPR 2018

3D Instance Segmentation

Input: 3D point cloud

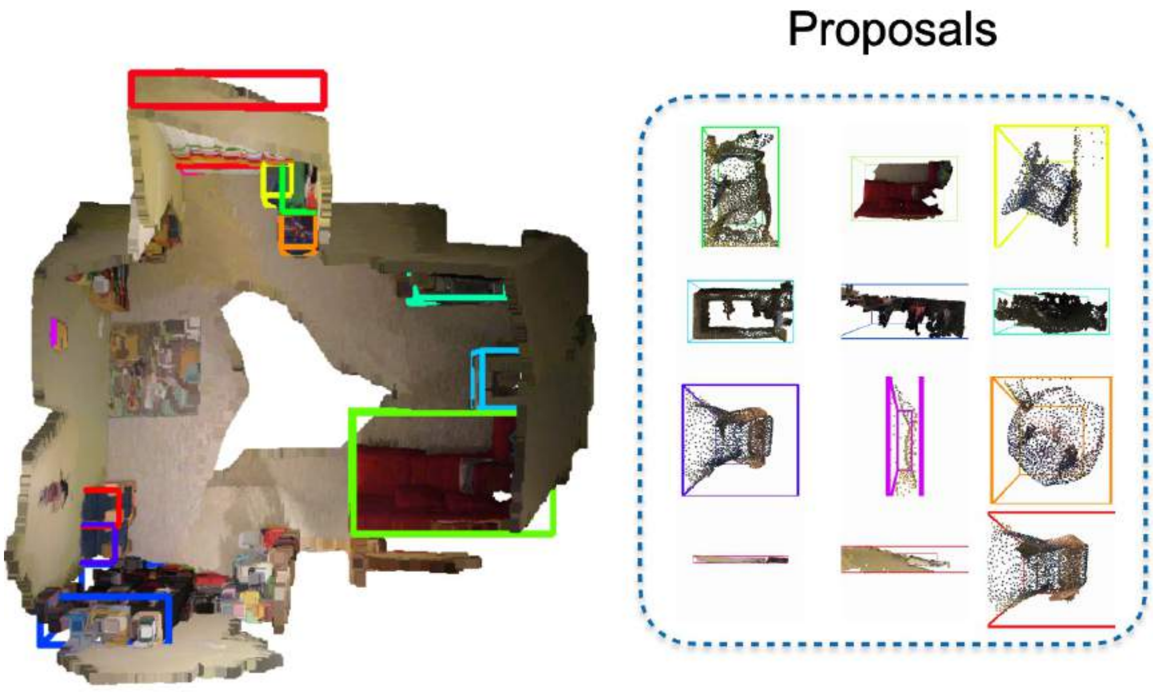
Output: instance labels of 3D points



3D Instance Segmentation

Top-down approach:

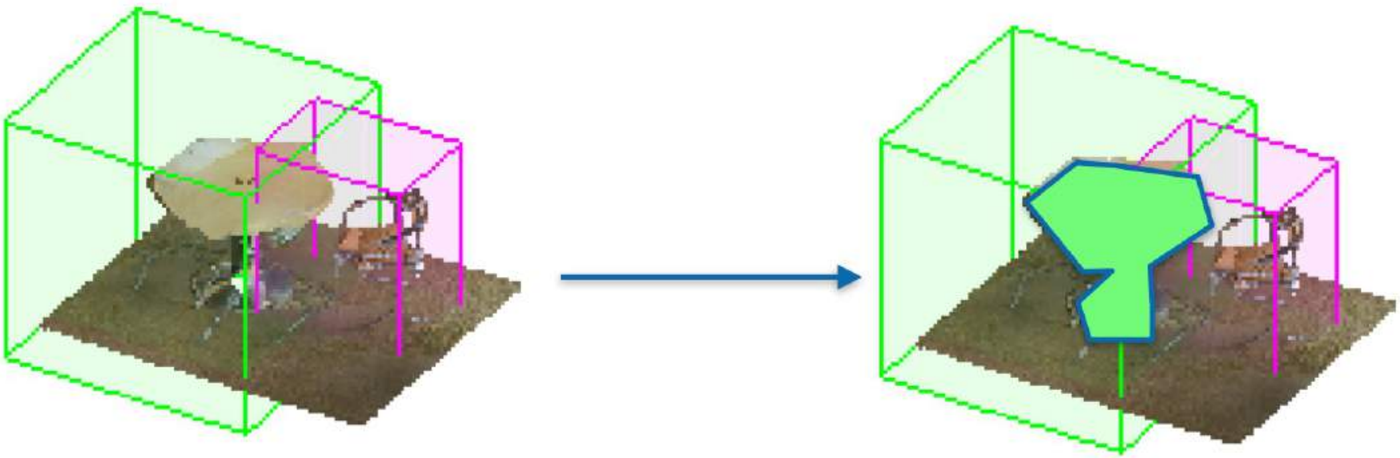
1. Run 3D detection



3D Instance Segmentation

Top-down approach:

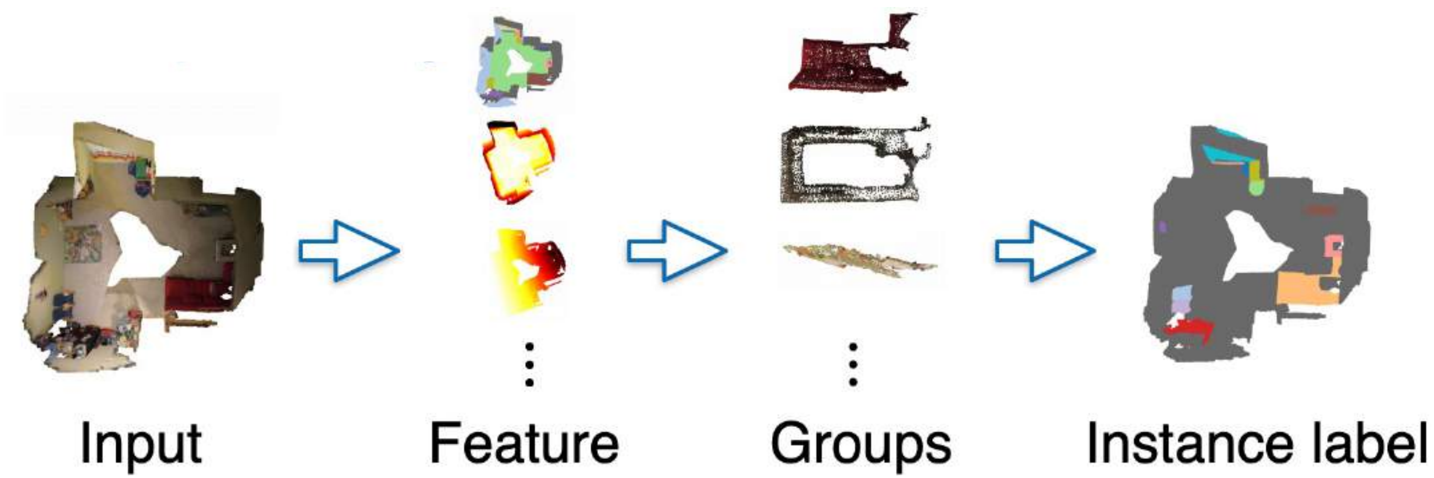
2. Run segmentation in each 3D bbox



3D Instance Segmentation

Bottom-up approach:

Group (cluster) points into different objects



Datasets for 3D Objects

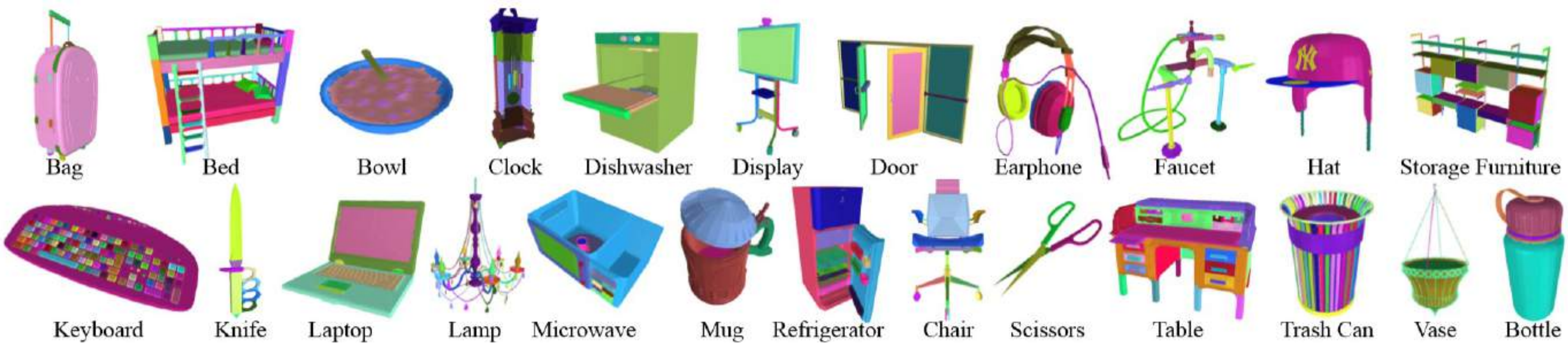
Large-scale Synthetic Objects: ShapeNet

Chang et al., “**ShapeNet: An Information-Rich 3D Model Repository**” , *arXiv*
Wu et al., "**3D ShapeNets: A deep representation for volumetric shapes**", *CVPR 2015*
Choi et al., “**A Large Dataset of Object Scans**", *arXiv*

Datasets for 3D Object Parts

Fine-grained Part: PartNet (ShapeNetPart2019)

Fine-grained (towards mobility)
Instance-level
Hierarchical



Mo et al., "**PartNet: A Large-Scale Benchmark for Fine-Grained and Hierarchical Part-Level 3D Object Understanding** ", *CVPR 2019*

Datasets for Indoor 3D Scenes

Large-scale Synthetic Scenes: SceneNet

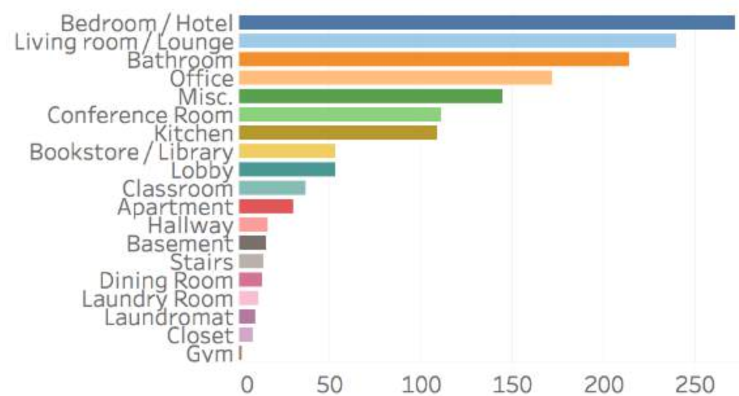
3D meshes
5M Photorealistic Images

Ankur et al., "**Understanding RealWorld Indoor Scenes with Synthetic Data**", *CVPR 2016*
McCormac et al., "**SceneNet RGB-D: Can 5M Synthetic Images Beat Generic ImageNet Pre-training on Indoor Segmentation?**", *ICCV 2017*

Datasets for Indoor 3D Scenes

Large-scale Scanned Real Scenes: ScanNet

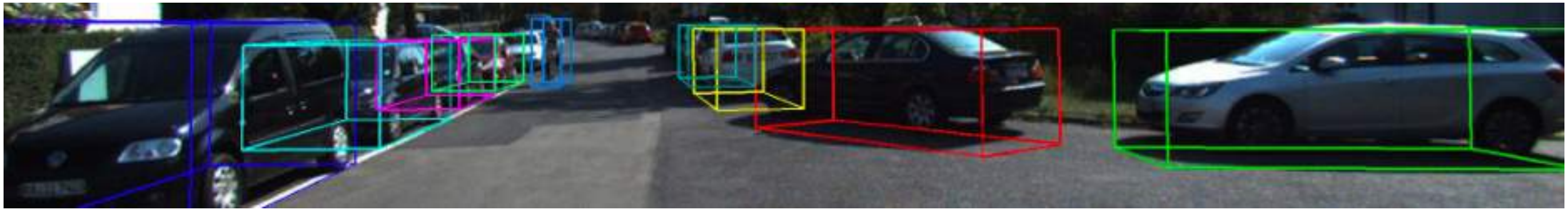
2.5 M Views in 1500 RGBD scans
3D camera poses
surface reconstructions
Instance-level semantic segmentations



Dai et al., “**ScanNet: Richly-annotated 3D Reconstructions of Indoor Scenes**”, *CVPR 2017*

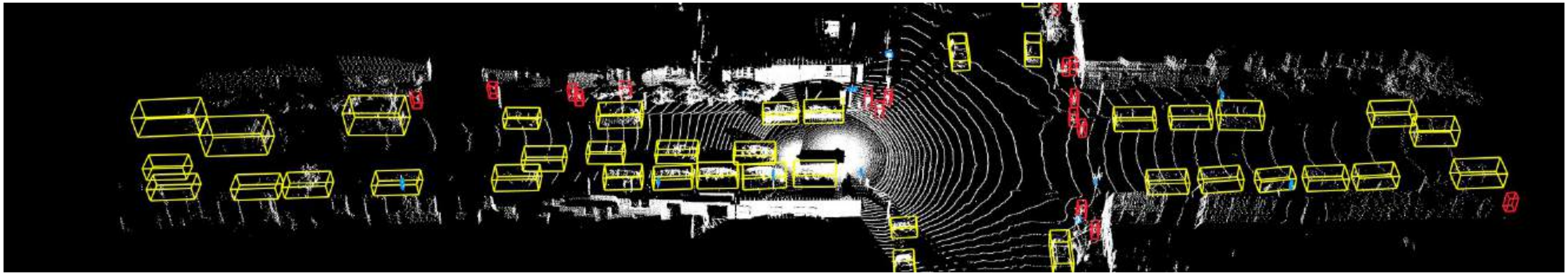
Datasets for Outdoor 3D Scenes

KITTI: LiDAR data, labeled by 3D bboxes



Semantic KITTI: LiDAR data, labeled per point

Waymo Open Dataset: LiDAR data, labeled by 3D b.boxes



Questions?